



FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

COMPUTER SCIENCE DEPARTMENT

CS0025 (SOFTWARE ENGINEERING 1)

ACTIVITY 1 [AI – ASSISTED]

FEASIBILITY STUDY FOR THE TARGET THESIS TITLE PROPOSAL

Group Name:	Statisfied (Section TS31)
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Date Performed:	September 14, 2026
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Assessment Purpose

To evaluate five target thesis titles and determine which proposal is the most feasible based on technology, algorithms, dataset availability, operations, cost, schedule, and ethical requirements. Each title is assessed separately against the ten required contents, then compared and ranked, and a primary recommendation and backup title are identified.

Titles Under Assessment

#	Plan	Target Thesis Title	Specialization	Primary Technique
1	A	Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost	Data Science	Predictive Analytics
2	B	Short-Term Luzon Grid Power Demand and Heat-Stress Forecasting Using LightGBM and Heat Index Telemetry	Data Science	Predictive Analytics
3	C	A Rule-Based and Isolation Forest Data Quality Remediation System for Barangay Tatalon Administrative Records	Data Science	Data Governance and Ethics
4	D	A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity	Data Science	Data Governance and Ethics
5	E	Retail Product Demand Forecasting and Price Optimization for Small and Medium Stores Using XGBoost	Data Science	Predictive Analytics



TARGET TITLE 1 OF 5: PLAN A

Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost

1. Target Thesis Title

Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost

Thesis specialization: Data Science

Primary technique: Predictive Analytics, from the approved Data Science technique list

Research domain: Agricultural market forecasting and decision support

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Benguet's highland farms supply temperate vegetables through the Benguet Agri-Pinoy Trading Center in La Trinidad. BAPTC brings farmers and traders together, but its publicly visible price information is fragmented and mainly backward-looking. Farmers therefore lack a reliable signal for planning harvests before overlapping supply causes sharp price declines.

Specific problem. There is no accessible short-term forecast of wholesale price movement or supply-glut risk by commodity for this market. Planting and harvest decisions are made without any forward indicator, and when many farmers harvest the same commodity in the same window, the resulting oversupply drives prices below recovery cost.

Target users and beneficiaries. The primary users are BAPTC and Department of Agriculture Cordillera Administrative Region staff. Farmers and traders are the intended beneficiaries, reached through existing DA-CAR communication channels rather than through individual accounts in the system.

Proposed solution. A predictive analytics system that forecasts short-term wholesale price movement and supply-glut risk by commodity, delivered as a web dashboard presenting forecasts, risk alerts, model performance, and downloadable summaries.

Main objectives.

- Confirm and obtain structured historical BAPTC price and trading-volume records.
- Use DA Bantay Presyo bulletins as an independent secondary signal for cross-validation.
- Engineer commodity, seasonal, lagged-price, lagged-volume, and rolling-trend features.
- Compare time-series and machine-learning models for price forecasting and glut-risk classification.
- Provide an early-warning dashboard for staff decision support.

Expected output. A cleaned historical price and volume dataset, evaluated forecasting and classification models, a web dashboard with forecasts and risk alerts, and a documented model performance report by commodity.

Intended contribution. An evaluated forecasting method for a Philippine highland-vegetable market that has not yet been modelled in a publicly identified study. The contribution depends on confirming access to suitable BAPTC records.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
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Programming languages	Python; SQL; JavaScript only if required	Python supports data preparation, forecasting, evaluation, and dashboard services in one stack.
Frameworks and libraries	pandas, NumPy, statsmodels, scikit-learn, XGBoost or LightGBM	All are widely used and open source. Simpler scikit-learn models can replace the boosted libraries if tuning proves difficult.
Database and storage	PostgreSQL or SQLite, with CSV and XLSX import	PostgreSQL fits a shared deployment; SQLite supports an offline prototype with no server administration.
APIs and third-party services	None on the critical path. An API connection is used only if BAPTC exposes one.	This deliberately avoids making an undocumented API a critical dependency.
Platform	Streamlit, or a FastAPI web service with a browser dashboard	A Streamlit prototype reduces development time; FastAPI offers more control if integration is later required.
Hardware and equipment	Development laptops, with optional free-tier cloud hosting	No sensors or specialised equipment are required. Gradient boosting at this data scale runs comfortably on a student laptop.
Architecture and integration	File-based import first, then staging database, then feature engineering, then model training and evaluation, then dashboard	File import keeps the pipeline working regardless of whether BAPTC ever publishes a machine-readable feed.
Security and maintenance	Role-based access, encrypted transport, validated uploads, database backups, and audit logs. Commodity-level public outputs are kept separate from any restricted raw extract.	If the group's web-development capacity is thin, the first version uses Streamlit, SQLite, and scheduled file imports instead of a custom service.
Technical expertise	Python, pandas, and dashboard work are all in hand. Time-series validation discipline is the least familiar area.	Rolling-origin validation is a procedural requirement rather than an advanced technique, and is documented in standard forecasting references.
Limitations and alternatives	The application stack is not the constraint. Historical data availability and consistency are.	DA Bantay Presyo retail bulletins provide an independently available secondary series if BAPTC records are delayed.

Technical finding. The software stack is feasible with open-source tools and the group has the required skills. The main technical constraint is not the application stack but the availability and consistency of the historical data, treated in section 4.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	BAPTC aggregate daily or weekly wholesale price and trading volume by commodity.
Source, owner, and provider	BAPTC and DA-CAR. As of September 10, 2026 the BAPTC Facebook price page replied with an instruction to send a formal request letter and a one-page proposal to baptc.center@bsu.edu.ph , addressed to Dr. Ruth C. Diego, General Manager. This is the first verified, responsive data-request channel this title has had.
Public link	No downloadable BAPTC dataset or API is confirmed. The BAPTC portal at baptc.bsu.edu.ph returned a DNS resolution failure on September 9, 2026. BAPTC price posts are a reference point, not the planned acquisition method.
Data type and file format	Unknown until the data inventory. Preferred formats are CSV, XLSX, or a database export.
Estimated size	Depends on coverage and reporting frequency. The record count must clear the 10,000-row course floor and is unverified.
Features and attributes	Commodity, date, unit, wholesale price, trading volume, month or season, lagged values, and rolling statistics.



Labels and target variables	Short-term price or price change for the regression task, plus a glut-risk label defined by an adviser-approved price-drop threshold for the classification task.
Accessibility and collection	Formal aggregate-data request to baptc.center@bsu.edu.ph addressed to the General Manager, with DA-CAR AMAD (amad@car.da.gov.ph) as a parallel channel. A matched letter and proposal package already exists in the group's working files.
Permission and licensing	Document data ownership, allowed uses, retention, publication limits, and whether a data-sharing agreement is required, before any transfer.
Quality, completeness, and bias	Expected issues include missing dates, inconsistent commodity names or units, incomplete volume records, later revisions, and bias toward commodities or periods with better record-keeping.
Preprocessing and storage	Standardise dates, commodities, units, missing-value rules, outliers, and duplicates. Keep raw extracts outside the public repository, record checksums and provenance, restrict access, and publish only approved aggregate outputs.
Training, validation, testing	Chronological rolling-origin validation, never random splitting. A final holdout period stays untouched until model selection is complete.
Alternative data plan	Compile DA Bantay Presyo retail bulletins as an independently available secondary series. This supports cross-validation and a narrower price-only study, but cannot replace BAPTC volume data for supply-glut forecasting. Direct Facebook scraping is a last resort requiring adviser approval and a terms-of-service and ethics review.

Dataset finding. Feasibility is conditional. A responsive request channel now exists, which is a material improvement over the position recorded earlier in the term, but no structured record has yet been received. The title should not proceed unchanged unless structured BAPTC records, adequate historical coverage, and permission for research use are all confirmed early in the term.

5. Algorithm or Model Feasibility

Evaluation Area	Approach 1	Approach 2	Approach 3
Name of algorithm or approach	SARIMA	Prophet	XGBoost gradient-boosted trees
Purpose in the proposed project	Transparent seasonal price baseline	Trend and seasonal forecast with interpretable components	Multivariate price regression and glut-risk classification
Dataset requirements	Regular, sufficiently long univariate series	Regular series; tolerates some missing observations	Lagged price and volume plus calendar and commodity features
Expected performance	Strong when seasonality is stable	Useful baseline; components are easy to explain	Potentially strongest where nonlinear interactions and volume signals matter
Computational requirements	Low	Low to moderate	Moderate; comfortably feasible on student laptops
Advantages	Interpretable with established diagnostics	Fast setup; handles multiple seasonal effects	Flexible; supports mixed features and both regression and classification in one model family
Limitations	Weak against abrupt nonlinear shifts	Can smooth over shocks and is sensitive to tuning	Requires careful leakage control and additional work to explain
Evaluation metrics	MAE, RMSE, MAPE or sMAPE	MAE, RMSE, MAPE or sMAPE	MAE and RMSE for price; precision, recall, F1, and PR-AUC for glut risk
Implementation complexity	Moderate	Low to moderate	Moderate

Recommended approach. XGBoost is the primary model, because the study needs one method that handles both the price regression and the glut-risk classification while using lagged volume alongside price, which a univariate seasonal model cannot do. SARIMA is retained as the transparent seasonal baseline and Prophet as a secondary baseline. Model selection uses rolling-origin validation and every candidate is compared against a naive last-value and seasonal-naive forecast, so an apparently good error figure cannot pass unchallenged. If volume data proves unavailable, the study narrows to price forecasting rather than continuing to claim supply-glut prediction.

6. Operational Feasibility

Area	Assessment
Target users and stakeholders	BAPTC and DA-CAR staff are the direct users and the data gatekeepers. Farmers and traders are beneficiaries rather than system users.
Intended operating environment	Office or trading-centre use through a browser on standard computers with a stable internet connection. An offline or local-network deployment supports limited connectivity.
Availability of participants	A BAPTC or DA-CAR contact who can explain the records, authorise access, and review forecast interpretation must be confirmed. A responsive channel exists but no named participant has yet agreed.
User needs and benefits	A forward indicator of price movement and oversupply risk by commodity, which does not currently exist for this market in any accessible form.
Devices and facilities	Standard office computers and a browser. No procurement is required.
Usability and accessibility	Dashboard terms, charts, colour contrast, keyboard access, and mobile readability tested with intended staff users.
Training and support	A short user guide and an administrator procedure for importing data, rerunning models, checking failures, and backing up records.
Maintenance	Scheduled model monitoring, because crop patterns, reporting practices, and market behaviour change over time and a model fitted once will drift.
Adoption challenges	The main risk is that forecasts are read as guarantees. Managed by displaying confidence ranges, last-update dates, data gaps, and model limitations rather than presenting single numbers as certainties.

Operational finding. A staff-facing dashboard is practical if BAPTC or DA-CAR participates in data clarification and user testing. Without a partner contact the project can still demonstrate a prototype, but operational validation would be considerably weaker.

7. Economic Feasibility

The amounts below are planning allowances, not supplier quotations. They assume the group already owns suitable laptops and uses open-source software throughout.

Expense	Estimated Cost	Purpose	Possible Alternative
Software licences	PHP 0	Development and analytics tools	Use open-source software
APIs and subscriptions	PHP 0 to 1,200	Optional services or notifications	Use file imports and free tiers
Cloud hosting and storage	PHP 0 to 2,400	Pilot deployment and backups	Local or school-hosted deployment
Dataset acquisition or annotation	PHP 0	Aggregate records and cleaning	Request institutional data; clean manually
Hardware or equipment	PHP 0	Development and testing	Use existing laptops
Transportation and data collection	PHP 3,000	BAPTC or DA-CAR visits	Use scheduled calls and email where appropriate, since the request channel is remote



Testing and deployment	PHP 1,000	User testing and incidentals	Use remote testing
Maintenance	PHP 1,200	Backups or post-pilot hosting	Local archive and scheduled exports
Contingency expenses	PHP 1,500	Unexpected project expenses	Reduce optional hosting or travel
Total estimated cost	PHP 6,700 to 10,300	Working budget range	The lower bound uses free tiers throughout

Economic finding. The project can be low-cost because the proposed stack and secondary datasets are free. This is the most expensive of the four assessed titles, driven almost entirely by travel to La Trinidad, which is roughly a five-hour trip from Metro Manila and therefore not a routine visit. The group must confirm its actual budget, travel plan, and any institutional data-access fees before final approval.

8. Schedule Feasibility

Activity	Estimated Duration	Dependency or Control
Requirements gathering	Weeks 1 to 2	Includes the data-access feasibility gate
Literature and related-study review	Weeks 1 to 4	Continues throughout model selection
Dataset acquisition or data collection	Weeks 1 to 3	Escalate or narrow scope if records are not confirmed by Week 3
Data cleaning, annotation, and preprocessing	Weeks 3 to 5	Depends on the received format and quality
System design	Weeks 3 to 5	Data schema, architecture, and dashboard flow
Algorithm or model development	Weeks 5 to 9	Build baselines before complex models
System implementation	Weeks 6 to 11	Dashboard developed in parallel
Model training and validation	Weeks 9 to 12	Rolling-origin evaluation
Software and user testing	Weeks 11 to 14	Requires partner or representative users
Performance evaluation	Weeks 12 to 14	Compare models and error by commodity
Revision and improvement	Weeks 14 to 16	Correct defects and refine explanations
Documentation	Weeks 4 to 17	Maintain methods and provenance as work proceeds
Deployment and final presentation	Weeks 17 to 18	Subject to the course calendar

Activities that may cause delays. The largest risks are data authorisation, poor data quality on arrival, and unavailable partner testers. These are managed by a Week 3 decision point: continue with the full title if BAPTC price and volume data are confirmed, otherwise adopt an adviser-approved narrower price-forecasting scope using accessible DA Bantay Presyo data.

Schedule finding. The 18-week plan is achievable, but it holds less slack than the other titles because the fallback still depends on assembling a usable secondary series rather than switching to something already in hand.

9. Legal and Ethical Feasibility

Requirement	Planned Control
Data Privacy Act compliance	Apply data minimisation, purpose limitation, access control, secure storage, retention limits, and approved disposal. The study is designed around commodity-level aggregates and does not collect trader or farmer identities.
Informed consent	Not expected for aggregate historical records. Consent is obtained if user testing collects identifiable responses.
Research ethics approval	Request an institutional ethics determination rather than assuming the project is exempt. Confirm the applicable review category before data collection, bulletin retrieval, or user testing.



Confidentiality and anonymisation	Use commodity aggregates, remove identifiers, restrict raw data, and report only approved summaries. If BAPTC data contains personal, confidential, or transaction-level fields, those fields must be excluded or anonymised before transfer and analysis.
Intellectual property and copyright	Cite sources, document model and software licences, and respect BAPTC and DA ownership and publication terms.
Licences	All proposed libraries carry permissive open-source licences compatible with academic use.
Data ownership and sharing restrictions	Use written authorisation or an agreement stating access, reuse, retention, publication, and deletion conditions.
Cybersecurity	Encrypt transfers, validate uploads, protect credentials, maintain backups, and log administrative activity.
AI fairness, transparency, accountability, bias	Report error by commodity and season; disclose uncertainty, missing data, thresholds, and known bias. A model that performs well on average may perform poorly on exactly the commodities where a glut matters most, so per-commodity error is reported rather than a single headline figure.
Possible misuse or harm	A forecast acted on as a certainty could cause real financial loss to a farmer. Forecasts are labelled as decision support, unsupported public claims are restricted, and the dashboard never presents a predicted price as a guaranteed one.

Legal and ethical finding. The aggregate-data design lowers privacy risk substantially, but written data-use conditions and an institutional ethics determination remain necessary before the full study proceeds.

10. Risk Assessment and Feasibility Conclusion

Identified Risk	Likelihood	Impact	Mitigation Strategy
Dataset is unavailable or insufficient	Low	High	A verified request channel now exists. Send the formal letter and proposal early, set a Week 3 decision gate, and narrow to price forecasting if volume data is unavailable.
Volume data withheld while price data is released	Medium	High	The glut-risk claim is dropped and the study narrows to price forecasting. This is a scope change, not a failure, and is decided at the Week 3 gate.
Technology is difficult to implement	Low	Medium	Use an open-source Python stack and a Streamlit prototype before adding integration complexity.
Algorithm produces poor results	Medium	High	Use naive and seasonal baselines, rolling-origin validation, feature review, and commodity-specific error analysis. A model that cannot beat seasonal-naive is reported as such.
Required respondents are unavailable	Medium	Medium	Recruit a small staff sample early and prepare remote or representative usability testing.
Privacy or ethical concerns arise	Low	High	Request only aggregates, secure raw data, obtain an ethics determination, and document authorisation.
Project exceeds the budget	Low	Medium	Use open-source tools, free tiers, existing devices, and a capped travel allowance. Travel to La Trinidad is the dominant cost and can be reduced by using the remote request channel.
Project cannot be completed on time	Medium	High	Time-box data access, prioritise a minimum viable dashboard, and trigger the narrower backup scope at Week 3.

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. To conclude, the software, the algorithms, the operating environment, the cost, and the planned safeguards are all achievable for a student group, which means this title can be built with the skills and the equipment the group already has. The position on data has also improved since the first assessment, since as of September 10, 2026 the group holds a verified and responsive request channel to the BAPTC General Manager, which it did not have before. However, the primary data source is still the single condition that decides this title,



because a responsive channel is not yet the same thing as a received dataset. The title may proceed once the group confirms exportable historical BAPTC price and volume records, research-use permission, adequate coverage, and a realistic delivery date. Yet if those conditions are not met by Week 3, the scope should narrow to a price-only forecasting study using DA Bantay Presyo bulletins, with the supply-glut claim withdrawn.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	4	Open-source stack is practical and familiar
Dataset and data feasibility	4	A verified and responsive request channel to the BAPTC General Manager; coverage and format still to be confirmed
Algorithm or model feasibility	4	Multiple established approaches with clear baselines available
Operational feasibility	4	A responsive institutional contact at BAPTC; the individual staff testers are named at the first meeting
Economic feasibility	4	Low-cost stack, but the highest travel cost of the four titles
Schedule feasibility	3	Feasible only with an early data decision gate
Legal and ethical feasibility	4	Aggregate design lowers risk; approvals are still required
Risk manageability	4	A named and responsive channel plus a defined fallback scope make this risk controllable
Total score	31/40	
Group's ranking	1	See the final comparison table

References

- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785-794.
- Department of Agriculture. (n.d.). BAPTC to spin into corporation to provide farmers, traders more services, bigger income. da.gov.ph
- Department of Agriculture Cordillera Administrative Region. (n.d.). Bantay Presyo price monitoring bulletins. car.da.gov.ph
- Freedom of Information Philippines. (n.d.). Program documents of the Benguet Agri-Pinoy Trading Center. foi.gov.ph
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and practice (3rd ed.). OTexts. otexts.com/fpp3/
- OpenGov Asia. (n.d.). The Philippines: Digital trading to transform Benguet's agri-business sector. opengovasia.com
- Philippine Information Agency. (n.d.). DA eyes digital trading, profiling at Benguet trade center. pia.gov.ph



TARGET TITLE 2 OF 5: PLAN B

Short-Term Luzon Grid Power Demand and Heat-Stress Forecasting Using LightGBM and Heat Index Telemetry

1. Target Thesis Title

Short-Term Luzon Grid Power Demand and Heat-Stress Forecasting Using LightGBM and Heat Index Telemetry

Thesis specialization: Data Science

Primary technique: Predictive Analytics, from the approved Data Science technique list

Research domain: Smart-grid analytics, time-series machine learning, energy economics, and urban environmental telemetry

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. The Luzon Grid supports the Philippines' largest population and commercial centres. During hot and humid periods, increased cooling use can create nonlinear demand surges that strain the grid and contribute to reserve alerts.

Specific problem. Public load-forecasting studies often use temperature alone, or assumptions designed for temperate climates where humidity matters less. Philippine grid-demand records are also fragmented and rarely aligned with localised hourly heat-index data, which limits the availability of accessible 24-hour forecasting tools grounded in local conditions.

Target users and beneficiaries. Grid and market analysts, distribution utilities, local energy planners, and academic researchers.

Proposed solution. Luzon GridWatch, a Python forecasting pipeline and Streamlit dashboard that combines public IEMOP and NGCP load records with hourly weather and calculated heat-index data to produce 24-hour demand forecasts and heat-stress visualisations.

Main objectives.

- Compile at least 10,000 aligned hourly grid and weather records.
- Create lag, calendar, and heat-index features from those records.
- Compare statistical and machine-learning models on the same chronological validation scheme.
- Visualise forecasts and simulated heat-stress conditions in a decision-support dashboard.

Expected output. A cleaned time-series dataset, evaluated forecasting models, reproducible code, comparative MAPE and RMSE results, and an interactive decision-support dashboard.

Intended contribution. An open and locally relevant framework for studying the relationship between tropical heat index and short-term Luzon Grid demand, requiring no hardware and no personal consumer data.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
Programming languages	Python 3.10 or later for ingestion, preprocessing, modelling, and dashboard logic;	All are standard undergraduate skills already used by the group.



	SQL for queries; HTML and CSS for minor interface styling	
Frameworks and libraries	pandas, NumPy, scikit-learn, statsmodels, LightGBM or XGBoost, Plotly, and Streamlit. TensorFlow or PyTorch only if the optional sequence model is attempted.	All open source. If deep learning proves too demanding, SARIMAX and boosted trees are retained and the optional component is dropped without affecting the core result.
Database and storage	CSV and Parquet during research; SQLite or PostgreSQL for cleaned records, model results, and forecasts	Parquet keeps a multi-year hourly series compact enough to work with on a laptop.
APIs and third-party services	IEMOP and NGCP public portals, plus a historical weather source from OpenWeatherMap, Visual Crossing, Open-Meteo, ERA5, or PAGASA	Multiple interchangeable weather providers exist, so no single service is a single point of failure. Open-Meteo and ERA5 require no paid key.
Platform	Browser-based Streamlit application hosted locally or on a free-tier service	If hosting fails, the system is demonstrated locally with no loss of evaluated result.
Hardware and equipment	Existing laptop or desktop with at least 8 GB RAM. Google Colab may support optional deep-learning experiments. No sensors are required.	LightGBM on a dataset of this size trains in seconds to minutes on a CPU.
Architecture and integration	Extract-load-transform pipeline, then feature store, then trained models, then forecast module, then dashboard	Cached raw downloads mean a source outage does not halt development.
Security and maintenance	Keep API keys in environment variables, exclude raw data from public repositories, record checksums, validate source schemas, back up models, and document retraining	No personal data is handled at any point, which removes an entire category of security obligation.
Technical expertise	The stack uses common undergraduate Python and data-science skills already demonstrated by the group.	The least familiar area is walk-forward validation discipline, which is procedural rather than advanced.
Limitations and alternatives	Grid reports may arrive as PDF-derived tables rather than clean CSV, and public file formats may change mid-study.	Cached raw files, documented parsers, and fallback weather sources absorb both problems.

Technical finding. Every component is familiar, open source, and runs on hardware the group already owns. No permission is needed from anyone to begin. This is the lowest-dependency technical position of the four assessed titles.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	Hourly Luzon Grid system demand in MW, merged with localised temperature, relative humidity, dew point, and calculated heat index.
Source, owner, and provider	Independent Electricity Market Operator of the Philippines, National Grid Corporation of the Philippines, and an approved historical weather provider.
Public link	iemop.ph; ngcp.ph; openweathermap.org/api; open-meteo.com. All are publicly reachable and none requires an institutional request letter.
Data type and file format	CSV, Excel, or PDF-derived grid reports, and JSON or CSV weather responses.
Estimated size	Target is 10,000 or more aligned hourly observations covering roughly 14 to 24 months. At hourly resolution, 14 months alone yields over 10,000 rows, so the course volume floor is cleared by arithmetic rather than by hope.
Features and attributes	Timestamp, system demand, temperature, humidity, heat index, dew point, pressure and cloud variables, hour, weekday, month, holiday flag, and demand and weather lags.
Labels and target	Hourly system demand over a 24-hour forecast horizon. A secondary grid-stress indicator may



variables	be derived from clearly documented alert thresholds.
Accessibility and collection	Download official public files and use Python scripts to retrieve and align historical weather by timestamp and location. No permission letter, no gatekeeper, and no waiting period.
Permission and licensing	Use only records allowed by official terms and cite all providers. No household, customer, or personal data is collected at any point.
Quality, completeness, and bias	Possible missing intervals, time-zone mismatches, changing file formats, reporting delays, weather-station location bias, and unusual typhoon or outage effects that do not represent normal demand behaviour.
Preprocessing and storage	Normalise timestamps and units, remove duplicates, document minor-gap imputation, flag anomalous events, calculate heat index, build lags, and preserve raw files with checksums.
Training, validation, testing	Chronological training, validation, and test periods with expanding-window walk-forward validation. Random splits are never used, because they leak future information into training.
Alternative data plan	Use Open-Meteo or ERA5 if the primary weather source fails; narrow to a continuous one-year or dry-season window if coverage is patchy; if grid logs fail the Week 1 audit, substitute a confirmed open power-system dataset.

Dataset finding. This is the strongest dataset position among the assessed titles and the only one the group can prove today without anyone's approval. The remaining work is a Week 1 audit to confirm that the public files actually yield 10,000 usable, time-aligned hourly records, which is a task the group performs itself rather than waits on.

5. Algorithm or Model Feasibility

Evaluation Area	Approach 1	Approach 2	Approach 3
Name of algorithm or approach	SARIMAX	LightGBM gradient-boosted trees	LSTM or GRU recurrent network
Purpose in the proposed project	Statistical baseline with weather inputs	Primary nonlinear 24-hour load model	Optional sequence-model comparison
Dataset requirements	Ordered hourly load and exogenous variables	10,000 or more rows with calendar, heat, and lag features	Normalised sequential windows; benefits from considerably more data
Expected performance	Moderate and interpretable	Strong tabular accuracy with fast training	Potentially strong but more variable across runs
Computational requirements	Low to moderate	Moderate; CPU is sufficient	High; a GPU is helpful and Colab would be needed
Advantages	Transparent and well established	Captures nonlinear heat-load effects and reports feature importance, which is direct evidence for the study's central claim	Learns long temporal dependencies
Limitations	May miss sudden nonlinear surges, which are precisely the events of interest	Needs tuning and careful leakage control	Less interpretable and prone to overfitting at this data volume
Evaluation metrics	MAPE and RMSE	MAPE and RMSE	MAPE and RMSE
Implementation complexity	Medium	Medium	High

Recommended approach. LightGBM is the primary model, because it handles nonlinear heat-load relationships, works well with engineered tabular time-series features, trains efficiently on available hardware, and produces feature-importance evidence that speaks directly to the research question about heat index and demand. SARIMAX is the required statistical baseline. The LSTM or GRU comparison is deliberately optional rather than promised,



because the available data volume sits at the low end of what a sequence model needs, and committing to it would put a component the group may not complete on the critical path.

6. Operational Feasibility

Area	Assessment
Target users and stakeholders	NGCP and IEMOP analysts, distribution utilities, energy planners, researchers, faculty, and student evaluators.
Intended operating environment	A modern web browser, an internet connection for live data or deployment, and an ordinary laptop or school workstation.
Availability of participants	No external participant is required for the core evaluation, which is the decisive operational advantage of this title. Faculty and student evaluators cover usability testing.
User needs and benefits	24-hour demand estimates, peak-load visualisation, comparison across heat conditions, and a reproducible analysis others can rerun.
Devices and facilities	Ordinary laptop with at least 8 GB RAM. No procurement, no site visit, and no partner facility required.
Usability and accessibility	Clear units and labels, readable colour choices, filters, downloadable forecasts, stated model limitations, and keyboard-operable controls where practical.
Training and support	A short user guide, a dashboard walkthrough, and administrator documentation for data updates and model retraining.
Maintenance	Monitor source formats and API limits, validate new records, version models, update dependencies, and retain backups.
Adoption challenges	Limited access to utility partners, analyst trust in student-built forecasts, and changing public data formats. Managed by using public records for core testing, disclosing uncertainty plainly, and keeping fallback connectors ready.

Operational finding. The only title among the four whose core evaluation depends on no external organisation. A utility partner would strengthen validation, but its absence does not block the study or reduce the deliverable.

7. Economic Feasibility

Expense	Estimated Cost	Purpose	Possible Alternative
Software licences	PHP 0	Python and open-source libraries	Free academic and open-source tools throughout
APIs and subscriptions	PHP 0 to 1,500	Historical weather access	Free-tier API, Open-Meteo, ERA5, or PAGASA at no cost
Cloud hosting and storage	PHP 0 to 1,000	Dashboard and file hosting	Streamlit Community Cloud or a local demonstration
Dataset acquisition or annotation	PHP 0	Public records; no annotation required	Official open archives
Hardware or equipment	PHP 0	Existing laptop, with optional Colab for the deep-learning experiment	School laboratory computer
Transportation and data collection	PHP 0	None required; all data is downloaded	Not applicable, since there is no site visit
Testing and deployment	PHP 0 to 1,000	Meetings and pilot testing	Online evaluation and local deployment
Maintenance	PHP 0	Handled within the study period	Not applicable during the thesis term
Contingency expenses	PHP 1,000	Unexpected service need	Reduce sites or narrow the time



			window
Total estimated cost	PHP 1,000 to 5,000	Working budget range	A fully free stack keeps the cost near PHP 1,000

Economic finding. The cheapest title of the four, and the only one with no transportation cost at all, because every data source is downloaded rather than collected in person.

8. Schedule Feasibility

Activity	Estimated Duration	Dependency or Control
Requirements gathering	Weeks 1 to 2	Includes the source audit and a Go or Conditional-Go decision
Literature and related-study review	Weeks 1 to 4	Load-forecasting literature is mature and openly accessible
Dataset acquisition or data collection	Weeks 2 to 4	Week 1 audit confirms that 10,000 usable aligned records exist before anything is built on the assumption
Data cleaning, annotation, and preprocessing	Weeks 2 to 4	Timestamp alignment and quality report
System design	Weeks 4 to 6	Feature engineering and storage setup
Algorithm or model development	Weeks 4 to 10	Naive forecast, then SARIMAX baseline, then LightGBM tuning, then the optional sequence model
System implementation	Weeks 10 to 13	Dashboard implementation
Model training and validation	Weeks 6 to 12	Walk-forward validation throughout
Software and user testing	Weeks 10 to 13	Usability evaluation with available testers
Performance evaluation	Weeks 10 to 13	Comparative MAPE and RMSE analysis
Revision and improvement	Weeks 14 to 16	Defect correction and refinement
Documentation	Weeks 2 to 16	Maintained continuously
Deployment and final presentation	Weeks 14 to 16	Final report and presentation preparation

Activities that may cause delays. Data gaps, changing file formats, API rate limits, and long hyperparameter tuning runs are the realistic delay sources. These are managed through the Week 1 feasibility gate, weekly milestones, cached raw data, named fallback sources, building baseline models early, and keeping the deep-learning component optional rather than mandatory.

Schedule finding. The most comfortable schedule of the four, because no activity on the critical path waits on a third party to respond.

9. Legal and Ethical Feasibility

Requirement	Planned Control
Data Privacy Act compliance	The study uses aggregated public grid and weather data, not personal or household records. Republic Act No. 10173 remains applicable to any usability-test information collected from participants.
Informed consent	Obtained for usability testing, with responses anonymised. Not applicable to the grid and weather data itself.
Research ethics approval	The dataset work is expected to be Exempt or minimal risk, subject to institutional confirmation. Any human-participant testing follows school approval rules.
Confidentiality and anonymisation	No household, customer, or consumer-level data is collected at any point, so there is nothing to anonymise in the core dataset.
Intellectual property and	Follow IEMOP, NGCP, weather-provider, software, and dataset terms; cite sources; do not



copyright	redistribute restricted raw files; retain attribution in code and documentation.
Licences	All proposed libraries carry permissive open-source licences compatible with academic use.
Data ownership and sharing restrictions	Public data used under its published terms. Raw files are excluded from public repositories.
Cybersecurity	Keep API keys in environment variables, limit file access, use backups and checksums, patch dependencies, and never expose credentials in repositories or in the dashboard.
AI fairness, transparency, accountability, bias	Report MAPE and RMSE by season or relevant period rather than as a single figure, document features and exclusions, use walk-forward validation, and disclose geographic and event-related bias. Weather-station location bias is stated plainly, since a station that does not represent the load centre will flatter the model.
Possible misuse or harm	The dashboard is labelled an academic planning aid, not an official NGCP dispatch or safety system. Uncertain forecasts are never presented as guaranteed operational instructions, because a grid operator acting on a student forecast would be a genuine safety concern.

Legal and ethical finding. The lowest-risk title of the four by a clear margin, because no personal data is collected at any stage. The only real ethical obligation is preventing the dashboard from being mistaken for an operational grid tool.

10. Risk Assessment and Feasibility Conclusion

Identified Risk	Likelihood	Impact	Mitigation Strategy
Dataset is unavailable or insufficient	Medium	High	Audit the sources in Week 1; use minor-gap imputation only where justified; narrow the period or pivot to a confirmed open power-system dataset.
Grid reports are PDF-only or change format	Medium	Medium	Cache raw files, document parsers, and keep fallback sources ready.
Technology is difficult to implement	Low	Medium	Use a familiar Python stack, begin with baselines, and keep deep learning optional.
Algorithm produces poor results	Medium	High	Compare naive, SARIMAX, and LightGBM on identical walk-forward splits, tune carefully, and analyse error by season and hour.
Required respondents are unavailable	Low	Low	Core evaluation uses public data only. Usability testing proceeds with faculty and students if industry staff are unavailable.
Privacy or ethical concerns arise	Low	Low	Aggregated public data, consent for user tests, anonymised responses, secure storage, and documented licences.
Project exceeds the budget	Low	Low	Open-source tools, free data and hosting, existing computers, and no travel cost.
Project cannot be completed on time	Low	High	Weekly milestones, early data verification, priority on LightGBM and a minimum viable dashboard, and the optional sequence model dropped if time runs short.

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. The software, skills, hardware, cost, and schedule are all manageable, and this title carries a decisive structural advantage over the other three: it depends on no external organisation granting anything. Every data source is publicly downloadable, the volume floor is cleared by arithmetic at hourly resolution, and no personal data is handled at any stage. The single condition is a Week 1 audit confirming at least 10,000 usable, time-aligned hourly grid and weather records, which is work the group performs itself. If the public sources fail that gate, the team narrows the period or activates the documented fallback dataset plan.

Provisional Feasibility Score for This Title



Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	5	Entirely familiar open-source stack; no permission needed to begin
Dataset and data feasibility	4	Public and downloadable today; a Week 1 audit is still needed to confirm usable aligned volume
Algorithm or model feasibility	5	Load forecasting is a mature problem with well-established baselines
Operational feasibility	4	No external participant required for the core evaluation
Economic feasibility	5	PHP 1,000 to 5,000 with no transportation cost
Schedule feasibility	4	No critical-path activity waits on a third party
Legal and ethical feasibility	5	No personal data collected at any stage of the study
Risk manageability	4	Every major risk is inside the group's own control, with named fallbacks
Total score	36/40	The highest-scoring title assessed
Group's ranking	2	See the final comparison table

References

Department of Energy Philippines. (n.d.). Power statistics and supply-demand outlook. doe.gov.ph

Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and practice (3rd ed.). OTexts. otexts.com/fpp3/

Independent Electricity Market Operator of the Philippines. (n.d.). WESM market reports and historical market data. iemop.ph

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30, 3146-3154.

National Grid Corporation of the Philippines. (n.d.). System operations and grid information. ngcp.ph

National Weather Service. (n.d.). The heat index equation. wpc.ncep.noaa.gov/html/heatindex_equation.shtml

OpenWeather. (n.d.). Historical weather data API documentation. openweathermap.org/api



TARGET TITLE 3 OF 5: PLAN C

A Rule-Based and Isolation Forest Data Quality Remediation System for Barangay Tatalon Administrative Records

1. Target Thesis Title

A Rule-Based and Isolation Forest Data Quality Remediation System for Barangay Tatalon Administrative Records

Thesis specialization: Data Science

Primary technique: Data Governance and Ethics, from the approved Data Science technique list

Research domain: Administrative data quality, local government records management, and public-sector data governance

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Barangay administrative records, meaning clearances, certificates, and locally issued permits, are the raw material for local reports and planning decisions. Those records are typically produced across several years, by different encoders, under changing forms and category conventions, and without a documented standard for what a complete or valid entry looks like. Their fitness for use is therefore rarely measured in a repeatable way. The official Quezon City directory reports 63,129 residents and 11,800 households in Tatalon, which establishes the barangay as a substantial service area but says nothing about how many administrative transactions exist.

Specific problem. Records may contain blank required fields, several labels for the same service, impossible dates, inconsistent fee values, duplicate transactions, or formats that change across years. Where corrections do happen, they are applied manually and without an audit trail, so an original value cannot be distinguished from a corrected one. Any demand analysis or official summary built on those records may then be misleading without anyone knowing.

Target users and beneficiaries. The direct users are the Barangay Tatalon records custodian and administrative staff who prepare recurring summaries. The beneficiaries are the barangay council, which relies on those summaries for planning, and the residents served by the decisions made from them.

Proposed solution. A web-based data quality system that profiles a non-identifying extract of barangay transaction records, applies a versioned rule catalogue encoding the barangay's own service dictionary and fee schedule, flags statistical anomalies that pass the rules using Isolation Forest, and produces a corrected derivative dataset with a complete and reversible change log. The source extract is never modified.

Main objectives.

- Define a field dictionary and a versioned rule catalogue for the selected records.
- Measure completeness, validity, consistency, uniqueness, and timeliness by field, service type, and period.
- Detect and classify record-level violations without modifying the source extract.
- Produce a corrected derivative dataset with a complete change log and reversible transformations.
- Evaluate detection performance using controlled error injection with known ground truth.
- Present recurring error patterns and source-level prevention recommendations in a dashboard.



Expected output. A rule catalogue specific to barangay records, a working quality and remediation system, a corrected derivative dataset with its audit trail, a measured detection-performance report, and a quality dashboard with an exportable findings report.

Intended contribution. A documented and evaluated data quality rule taxonomy for Philippine barangay administrative records, which does not currently exist in published form, together with a controlled validation study that reports detection performance against known ground truth rather than asserting that the cleaning worked.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
Programming languages	Python 3.11 for profiling, rules, remediation, and dashboard logic; SQL for staging queries; minor HTML and CSS for interface styling	All are standard undergraduate skills already used by the group.
Frameworks and libraries	pandas and NumPy for profiling; Great Expectations or a custom assertion layer for the rule engine; scikit-learn for Isolation Forest; RapidFuzz for duplicate detection; Plotly for charts; Streamlit for the dashboard	All open source and installable without licences. A custom pandas assertion layer replaces Great Expectations if its configuration overhead proves heavy.
Database and storage	SQLite for the immutable staging table and derivative versions during development; PostgreSQL if a shared deployment is later required; CSV, XLSX, or database export accepted on import	SQLite needs no server and suits a single-workstation study. The design is portable to PostgreSQL without schema changes.
APIs and third-party services	None required. No external API is on the critical path.	This is a deliberate design choice, since it removes rate limits, key management, and third-party outages as project risks.
Platform	Browser-based application run locally, or on a free-tier host only if the dataset used for the demonstration is synthetic	The restricted real extract is never hosted externally.
Hardware and equipment	Existing group laptops with 8 GB RAM. No sensors, no specialised equipment, no GPU.	Isolation Forest on a dataset of this size trains in seconds on a CPU.
Architecture and integration	Restricted raw file, then file validation and checksum, then schema mapper, then immutable staging table, then profiler and rule engine, then violation register, then remediation engine, then corrected derivative with change log, then dashboard	File-based import only. The barangay production system is never connected to or modified, which is both a technical simplification and an ethics requirement.
Security and maintenance	Raw and corrected extracts stored outside Git and outside the thesis vault; checksums and provenance recorded on import; prohibited-field scan before any analytical module loads data; role-based access and audit logging in the application; scheduled backups	The prohibited-field scan is a hard gate, so a file containing excluded identifying fields cannot be processed even by mistake.
Technical expertise	The group has working Python, pandas, SQL, and Streamlit experience. Isolation Forest is a single scikit-learn estimator with few parameters.	The heaviest unfamiliar component is reversible transformation design, which is an engineering discipline rather than an advanced algorithm.
Limitations and alternatives	The rule catalogue cannot be written until the barangay's own service dictionary and fee schedule are seen. Until then, rules are drafted against a synthetic schema.	A synthetic barangay dataset built from published QC service categories lets development and controlled error-injection evaluation proceed before the real extract arrives.



Technical finding. The stack is straightforward and fully open source, and there is no component on the critical path that the group has not already used. The technical risk on this title is the lowest of the five. The real constraint lies entirely in data access, treated in section 4.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	Barangay Tatalon administrative transaction records covering clearances, certificates, and locally issued permits, requested as originally encoded rather than pre-cleaned.
Source, owner, and provider	Barangay Tatalon, Quezon City. The records custodian and the signatory for a cooperation letter are both unidentified as of this assessment.
Public link	None. No barangay-level transaction dataset is published anywhere. The Quezon City barangay directory page for Tatalon provides population and household counts only, not transaction data.
Data type and file format	Unknown. Preferred formats are CSV, XLSX, or a database export. A PDF-only or paper-only answer is a fatal outcome for this title and is the single most important question in the feasibility gate.
Estimated size	Unknown. The study requires at least 10,000 usable transaction rows to meet the course volume floor. With 11,800 households, a multi-year clearance history plausibly clears that floor, but this is an estimate and not a verified count.
Features and attributes	Transaction date, service category, purpose category, purok or zone, fee, status, encoder or office code where available, and an optional barangay-generated pseudonymous reference.
Labels and target variables	For the rule engine, each record receives a pass or fail result per rule, so no pre-existing labels are needed. For the evaluation, ground-truth labels are generated by the group through controlled error injection into a benchmark copy, which means detection performance can be measured exactly.
Accessibility and collection	Courtesy visit first, with no file requested on that visit, to confirm the gate questions and identify the custodian. An adviser-reviewed formal request letter follows. A complete letter draft already exists in the group's working files.
Permission and licensing	Written cooperation or a dated confirmation of permitted scope is required before transfer. The letter must state that no data will be opened before written ethics clearance, that the raw extract will not be committed to any repository, that the production system will not be modified, and that a quality report will be returned to the barangay.
Quality, completeness, and bias	Expected issues include blank required fields, several labels for one service, impossible or transposed dates, inconsistent fee values, duplicates, and schema drift across years. Bias is expected toward recent years, since earlier records are more likely to be paper or partially digitised.
Preprocessing and storage	Immutable raw copy with checksum and provenance record; all work performed on versioned derivatives; prohibited-field scan before processing; encoding detection and schema mapping on import; no silent deletion at any stage.
Training, validation, testing	Not a supervised learning split. The evaluation design is a controlled benchmark: a clean subset is copied, known errors are injected at controlled rates, and precision, recall, and F1 are measured against that ground truth. Isolation Forest is fitted on the record feature matrix with contamination treated as a reported sensitivity parameter rather than a tuned result.
Alternative data plan	Three fallbacks in order. First, a different cooperating barangay if Tatalon declines, since the method is not Tatalon-specific. Second, a Quezon City citywide administrative dataset, which converges with Plan D. Third, a fully synthetic barangay records generator built from published QC service categories, which preserves the method and the controlled evaluation but loses the real-world finding and weakens the contribution substantially.

Dataset finding. This is the weakest area of the title, and it is the reason the title cannot yet be declared feasible as proposed. The access channel is open, since a contact within Barangay Tatalon has already been reached by message and by email and a courtesy visit is set to follow. What remains unconfirmed is the state of the records



themselves, which is whether they are already digital and whether they reach the volume the study needs. Those two answers cannot be obtained from the outside, which is why the courtesy visit carries the decision.

5. Algorithm or Model Feasibility

Evaluation Area	Approach 1	Approach 2	Approach 3
Name of algorithm or approach	Deterministic rule engine with versioned constraints	Isolation Forest	Fuzzy matching with blocking, using Levenshtein and token-set ratio
Purpose in the proposed project	Measure the five quality dimensions and flag violations that can be stated as explicit conditions	Flag records that pass every rule but are statistically unusual across combined fields, which is where undocumented error patterns hide	Detect near-duplicate transactions where the same entry was encoded twice with small spelling or spacing differences
Dataset requirements	The barangay service dictionary and fee schedule, so valid values can be declared. No labels needed.	A numeric and encoded feature matrix of at least a few thousand rows. No labels needed.	Text fields of reasonable consistency and a blocking key such as date or service type to keep comparisons tractable
Expected performance	Near-perfect on violations the rules describe, and blind to everything they do not	Moderate. Useful as a secondary signal; produces false positives that need review	Strong on exact and near-exact duplicates; degrades on heavily inconsistent free text
Computational requirements	Very low. Vectorised pandas operations.	Low. Trains in seconds on a CPU for datasets of this size.	Moderate. Quadratic without blocking, which is why blocking is mandatory.
Advantages	Fully transparent and auditable; every flag traces to a named rule and can be defended to the barangay	Finds error patterns nobody thought to write a rule for; requires no labels	Addresses the uniqueness dimension directly, which rules alone handle poorly
Limitations	Can only find what it was told to look for; the catalogue is only as good as the dictionary behind it	Not explainable by itself; an anomaly score is not a reason, so every flag needs human adjudication before any correction	Threshold choice is subjective and a wrong threshold merges genuinely distinct records
Evaluation metrics	Rule pass rate per dimension; precision, recall, and F1 against injected errors	Precision at k, recall against injected anomalies, and false-positive rate; contamination reported as a sensitivity analysis	Precision, recall, and F1 on injected duplicate pairs
Implementation complexity	Low to moderate. The effort is in catalogue design, not code.	Low. One scikit-learn estimator with few parameters.	Moderate. Blocking strategy and threshold calibration take the most time.

Recommended approach. The deterministic rule engine is the primary method, because a data governance deliverable must be defensible to the record owner and every flag has to trace to a stated rule. Isolation Forest is the named secondary algorithm, retained because rules alone cannot surface error patterns that were never anticipated, and because it supplies the specific-algorithm requirement of the thesis guidelines without turning a governance study into a black box. Fuzzy matching is included as a targeted module for the uniqueness dimension only. Isolation Forest output is treated as a review queue and never as an automatic correction. All three are evaluated on the same controlled error-injection benchmark, which is what allows this title to report measured performance rather than assert success.

6. Operational Feasibility

Area	Assessment
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Target users and stakeholders	Barangay Tatalon records custodian and administrative staff as direct users; the barangay council and the adviser and panel as secondary stakeholders.
Intended operating environment	A barangay office workstation or a group laptop running the application locally in a browser. No internet connection is required for core operation, which suits an office with unreliable connectivity.
Availability of participants	Open and already contacted. The study needs one custodian who can explain the forms, the service categories, and the fee schedule, and the courtesy visit is the step that identifies that person. Until then the rule catalogue is drafted from published QC service categories and confirmed against the barangay's own conventions afterwards.
User needs and benefits	A measured view of which fields and periods are fit for reporting, faster preparation of recurring summaries because categories and rules are documented, an audit trail that explains every correction, and a reusable method for later record batches.
Devices and facilities	Ordinary laptop or office desktop with 8 GB RAM, a modern browser, and a secure location for the restricted extract. No procurement required.
Usability and accessibility	Plain-language labels rather than data quality jargon, readable colour contrast, keyboard-operable controls, and a violation register that staff can filter without training. A quality score is never shown without its underlying dimension breakdown.
Training and support	A short user guide and an administrator procedure covering import, rule updates, reviewing the violation register, exporting the corrected file, and backups. One walkthrough session with the custodian.
Maintenance	Rule catalogue versioning as forms and fee schedules change, dependency updates, and backup of the audit log. The rule catalogue is the component that will date fastest and it is designed to be edited without touching code.
Adoption challenges	Staff may reasonably read a quality assessment as criticism of their work. The system is framed and presented as measuring the records, not the people, results are returned to the barangay before anything is written up, and a high measured quality score is treated as a valid and publishable result rather than a failed study.

Operational finding. The operating environment is undemanding and the deliverable is genuinely useful to a small office. Operational feasibility rests on working with one records custodian, and the group has already made contact with the barangay, so what the courtesy visit settles is which custodian handles the forms rather than whether anyone is available at all. The framing risk around staff perception is real and has been designed for rather than ignored.

7. Economic Feasibility

Expense	Estimated Cost	Purpose	Possible Alternative
Software licences	PHP 0	Python, pandas, scikit-learn, Streamlit, SQLite	All are open source; no paid tool is on the critical path
APIs and subscriptions	PHP 0	None required by design	Not applicable, since no external service is used
Cloud hosting and storage	PHP 0 to 600	Optional demonstration host for a synthetic dataset only	Local demonstration on a laptop; the real extract is never hosted
Dataset acquisition or annotation	PHP 0	Records requested at no cost; error injection is scripted	Manual annotation is not needed because ground truth is generated
Hardware or equipment	PHP 0	Existing group laptops	School laboratory workstation if a laptop fails
Transportation and data collection	PHP 2,400	Courtesy visit, letter submission, custodian walkthrough, and results return, estimated at four trips for two members	Combine the letter submission with the courtesy visit and hold follow-ups by phone
Printing and documents	PHP 500	Formal letters, the proposal packet,	Submit digitally where the barangay



		and the returned quality report	accepts it
Testing and deployment	PHP 600	Walkthrough session incidentals	Conduct the walkthrough during an already scheduled visit
Maintenance	PHP 0	Handled within the study period	Not applicable during the thesis term
Contingency expenses	PHP 1,000	Additional visits or a change of partner barangay	Reduce trips if the custodian prefers remote coordination
Total estimated cost	PHP 4,500 to 5,100	Working budget range	The lower bound applies if no optional hosting is used

Economic finding. This is the least expensive of the five titles. Open-source software and free records mean the only real spending is transportation and printing, both of which fall further if the barangay accepts digital submissions. Cost is not a constraint on this title.

8. Schedule Feasibility

Activity	Estimated Duration	Dependency or Control
Requirements gathering	Weeks 1 to 2	Includes the courtesy visit and the feasibility gate questions
Literature and related-study review	Weeks 1 to 5	Independent of data access, so it proceeds regardless of the gate outcome
Dataset acquisition or data collection	Weeks 1 to 4	Hard decision point at Week 4. If no written permission and no confirmed digital extract exist by then, switch to the fallback barangay or to the synthetic generator
Data cleaning, annotation, and preprocessing	Weeks 4 to 6	Depends on the received format; a PDF-derived extract adds roughly two weeks
System design	Weeks 3 to 5	Runs in parallel with acquisition using the synthetic schema, so no time is lost waiting
Algorithm or model development	Weeks 5 to 9	Rule engine first, then Isolation Forest, then the duplicate module
System implementation	Weeks 6 to 11	Dashboard and violation register built alongside the engine
Model training and validation	Weeks 9 to 12	Controlled error injection and performance measurement
Software and user testing	Weeks 11 to 14	Requires the custodian walkthrough; faculty and student testers substitute if unavailable
Performance evaluation	Weeks 12 to 14	Detection performance plus weight-sensitivity analysis
Revision and improvement	Weeks 14 to 16	Defect correction and rule catalogue refinement
Documentation	Weeks 2 to 17	Maintained continuously; provenance recorded as work proceeds
Deployment and final presentation	Weeks 17 to 18	Handover package only; no live deployment is permitted during the study

Activities that may cause delays. Barangay response time is the largest and it is entirely outside the group's control. A PDF-only or paper-only extract is the second, since manual transcription would consume weeks that the schedule does not hold. Custodian availability for the walkthrough is the third. These are managed by starting development against a synthetic schema in Week 3 so that no development time depends on the extract arriving, by setting a hard Week 4 decision point with two named fallbacks, and by keeping the rule catalogue editable as configuration so a late dictionary does not force code rewrites.

Schedule finding. The 18-week plan is achievable, but only because development is deliberately decoupled from data arrival. Without that decoupling the title would not fit the term.



9. Legal and Ethical Feasibility

Requirement	Planned Control
Data Privacy Act compliance	Republic Act No. 10173 applies. Data minimisation is enforced technically: a prohibited-field scan blocks processing of any file containing excluded identifying fields. Purpose limitation, access control, retention limits, and documented disposal are all specified before collection.
Informed consent	Not applicable to the records themselves, which are institutional and non-identifying by the time they are received. Informed consent is obtained for the custodian walkthrough if any identifiable response is recorded.
Research ethics approval	Exempt classification is targeted on the basis of no human participants and no identifiable private information. Appendix I applies for formally authorised institutional records, and Appendix D applies because the barangay is explicitly a community organisation and data gatekeeper. Adviser confirmation is still required on whether Appendix H applies to automated rule-based remediation. No data is collected before the official ethics decision.
Confidentiality and anonymisation	The Record of Barangay Inhabitants is excluded entirely. Names, exact addresses, contact details, birthdates, government identification numbers, photographs, health records, blotter narratives, and complaint narratives are excluded and must not be received. Small-cell suppression applies to any aggregate that could expose a rare case.
Intellectual property and copyright	All sources cited; library and dataset licences documented; the barangay retains ownership of its records and the group claims no right to redistribute them.
Licences	Every library on the stack carries a permissive open-source licence compatible with academic use.
Data ownership and sharing restrictions	Governed by the written cooperation terms. The raw and corrected extracts are kept outside Git and outside the thesis vault, and are destroyed within thirty days after the final defence unless the permission terms require earlier.
Cybersecurity	Restricted storage path, checksum verification on import, credentials kept in environment variables, backup of the audit log, and administrative activity logging.
AI fairness, transparency, accountability, bias	Isolation Forest output is advisory only and never triggers an automatic correction. Every correction records the source hash, rule version, before-and-after value, reason, and timestamp, and every transformation is reversible. Dimension scores are always shown alongside any overall score. Expected bias toward recent and better-digitised years is reported rather than silently carried.
Possible misuse or harm	The realistic harm is that a quality score is read as a judgement of individual staff performance. Controls: results are reported at field, service, and period level and never by encoder identity, findings are returned to the barangay before write-up, and the system makes no decision about any resident.

Legal and ethical finding. The exclusion list and the technical prohibited-field gate put this title on defensible ground, and the design targets Exempt deliberately rather than hoping for it. Two items remain genuinely open: the Appendix H question on automated remediation, and written cooperation terms that do not yet exist. Both must be settled before any data is received.

10. Risk Assessment and Feasibility Conclusion

Identified Risk	Likelihood	Impact	Mitigation Strategy
Dataset is unavailable or insufficient	Low	High	Courtesy visit in Week 1; hard Week 4 decision point; three named fallbacks in priority order, being an alternative barangay, a citywide QC dataset, then a synthetic generator. The method survives all three, though the contribution weakens with each step.
Records are paper-based or PDF-only	Medium	High	This is the first question asked at the courtesy visit, before any letter is drafted, so a fatal answer is discovered in Week 1 rather than Week 8.



Volume falls below the 10,000-record floor	Medium	High	Ask for the approximate row count at the courtesy visit; widen the year range or the service types included; combine with a second cooperating barangay if needed.
Technology is difficult to implement	Low	Low	The stack is entirely familiar, no external API is on the critical path, and no specialised hardware is needed.
Algorithm produces poor results	Low	Medium	Controlled error injection gives exact ground truth, so detection performance is measured rather than claimed. If Isolation Forest adds nothing over the rules, that is itself a reportable and defensible finding.
Required respondents are unavailable	Medium	Medium	Only one custodian is strictly required. Faculty and student testers substitute for usability evaluation, and the rule catalogue can be grounded in published QC service categories if the dictionary is not released.
Privacy or ethical concerns arise	Low	High	Hard exclusion list, technical prohibited-field gate, Exempt-targeted design, no deployment during the study, and destruction within thirty days after the defence.
Project exceeds the budget	Low	Low	Total planned spending is under PHP 5,100, almost all of it transportation, which falls further with digital submission.
Project cannot be completed on time	Medium	High	Development is decoupled from data arrival via the synthetic schema, the Week 4 gate forces an early decision, and the dashboard is scoped as a minimum viable product first.

Conclusion for this title: **FEASIBLE WITH MODIFICATIONS.**

Supporting findings. To conclude, the technology, the cost, the schedule, and the ethical design of this title are all sound, and on those four criteria it is the strongest among the group's titles, which is shown by a stack that is entirely familiar, a total cost under PHP 5,100, and a controlled error-injection evaluation that produces measured results rather than asserted ones. The group has also already opened the channel that this title depends on, since a contact within Barangay Tatalon has been reached by message and by email, and a courtesy visit is set to follow. However, two things remain unconfirmed, which are whether the barangay's transaction records are already digital and whether they reach the ten thousand row floor that the study needs. Neither of these can be answered from the outside, which is why the courtesy visit is treated as the real decision point and not as a formality. Yet if both are still unconfirmed by Week 4, the group should move to an alternative cooperating barangay, and only then to the synthetic generator, which preserves the method and the evaluation but costs the study its real-world finding.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	5	Familiar open-source stack, no external API dependency, no specialised hardware
Dataset and data feasibility	3	Channel open and outreach sent; record format and volume still unconfirmed
Algorithm or model feasibility	4	Established methods with exact ground truth available through controlled error injection
Operational feasibility	3	Undemanding environment; the custodian is identified at the courtesy visit
Economic feasibility	5	Under PHP 5,100, almost entirely transportation
Schedule feasibility	3	Fits 18 weeks only because development is decoupled from data arrival
Legal and ethical feasibility	3	Exempt-targeted with hard exclusions, but barangay records carry the



		highest inherent privacy exposure of the five titles
Risk manageability	3	Outreach is sent and three fallbacks are named in priority order, so no single answer ends the title
Total score	29/40	Provisional, pending the full five-title comparison
Group's ranking	5	See the final comparison table

References

- Batini, C., Cappiello, C., Fractalanci, C., & Maurino, A. (2009). Methodologies for data quality assessment and improvement. *ACM Computing Surveys*, 41(3), Article 16.
- Ehrlinger, L., & Woss, W. (2022). A survey of data quality measurement and monitoring tools. *Frontiers in Big Data*, 5, 850611.
- Government Data Quality Hub. (2020). The Government Data Quality Framework. gov.uk
- International Organization for Standardization. (2008). ISO/IEC 25012:2008 Software engineering: Software product Quality Requirements and Evaluation (SQuaRE): Data quality model.
- Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). Isolation Forest. *Eighth IEEE International Conference on Data Mining*, 413-422.
- National Privacy Commission. (2012). Republic Act No. 10173: Data Privacy Act of 2012. privacy.gov.ph
- Quezon City Government. (n.d.). Tatalon. quezoncity.gov.ph/brgy-directory/tatalon/
- Wang, R. Y., & Strong, D. M. (1996). Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems*, 12(4), 5-33.
- Reference details and the full academic literature review must be verified and completed before final submission.*



TARGET TITLE 4 OF 5: PLAN D

A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity

1. Target Thesis Title

A Data Governance System for Privacy Release Readiness of Quezon City Open Data Using Mondrian k-Anonymity

Thesis specialization: Data Science

Primary technique: Data Governance and Ethics, from the approved Data Science technique list

Research domain: Statistical disclosure control, open government data, and privacy-preserving data publishing

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Quezon City publishes an Open Data Dashboard through its Information Technology Development Department together with source departments, with listed dashboards containing data from 2019 onward. Current programmes include Helpline 122, QCitizen ID, the PWD community, maintenance medicine, Pangkabuhayang QC, and scholarships, and the city has stated its intent to publish more. The official ITDD page lists a service called Control of Data Services in Conformity with the Data Privacy Act, under which an approved letter is forwarded to the Database Management Division for extraction, which is direct evidence that ITDD is the correct office to approach.

Specific problem. Every release has to answer two questions at once: whether the source data is fit for public use, and how much detail can be published without making a person identifiable. Removing names does not settle the second question. A rare category combined with a small geography and a narrow date can single out a person even when no direct identifier remains. At the same time, aggregating everything to be safe can make a dataset too coarse to answer any useful question. At present there is no documented procedure for comparing candidate publication granularities in terms of both residual disclosure risk and analytical utility, and no auditable record of why a given date, geography, category, or suppression level was chosen.

Target users and beneficiaries. The direct users are ITDD staff and source-department data owners who decide what a release should look like. The beneficiaries are the residents whose records appear in those datasets, and the researchers, journalists, and civic groups who use QC open data and currently have no documented statement of what a published dataset can and cannot support.

Proposed solution. A web-based governance system that profiles an ITDD-approved de-identified extract, classifies every field as a direct identifier, quasi-identifier, sensitive attribute, or ordinary attribute, generates candidate release profiles by generalising dates, geography, and categories, measures residual disclosure risk and information loss for each profile using Mondrian k-anonymity, and produces an auditable release-readiness report recommending publication settings.

Main objectives.

- Define a release-readiness metadata profile for a selected QC Open Data source.
- Measure completeness, validity, consistency, uniqueness, and timeliness using versioned rules.
- Classify direct identifiers, quasi-identifiers, sensitive attributes, and non-identifying fields from the approved schema.
- Measure equivalence-class and small-cell risk under candidate publication granularities.
- Apply controlled generalisation and suppression without altering the restricted source copy.



- Quantify information loss and aggregate-query error produced by each strategy.
- Compare a fixed suppression baseline against Mondrian k-anonymity on the same risk threshold.
- Produce an auditable release-readiness report documenting assumptions and unresolved risks.

Expected output. A field governance classification for the selected dataset, a measured risk and utility comparison across candidate release profiles, a working release-readiness system with an aggregate dashboard, an exportable report, and a reusable configuration format adaptable to other QC datasets.

Intended contribution. An evaluated and reusable procedure for measuring the quality, risk, and utility trade-off of a specific Philippine local government open-data release. National Privacy Commission guidance establishes that de-identification is not automatically anonymisation, but no published work applies that standard as a measurable, auditable workflow to a Philippine city release.

3. Technical and Technology Feasibility

Area	Proposed Choice	Feasibility and Alternative
Programming languages	Python 3.11 for profiling, risk measurement, generalisation, and dashboard logic; SQL for staging queries; minor HTML and CSS for styling	Standard undergraduate skills already used by the group.
Frameworks and libraries	pandas and NumPy for profiling and equivalence-class computation; a custom Mondrian implementation, with ARX or pycanon as cross-check references; scikit-learn for encoding utilities; Plotly for risk and utility charts; Streamlit for the dashboard	Mondrian is roughly 200 lines of recursive partitioning code and is well documented in the literature, so a custom implementation is realistic and also makes the mechanism defensible at the defence.
Database and storage	SQLite for the immutable staging copy and release-profile versions; PostgreSQL if shared deployment is later required; CSV or database export accepted on import	No server administration needed during the study.
APIs and third-party services	None on the critical path. The QC Open Data Dashboard is inspected manually through the browser to record what the public surface exposes.	Nothing behind a login is scraped or automated, which is both a policy requirement and a removal of technical risk.
Platform	Browser-based application run locally. A free-tier host is used only for a synthetic or already-public demonstration dataset.	The restricted record-level extract is never hosted externally under any circumstance.
Hardware and equipment	Existing group laptops with 8 GB RAM. No sensors, no GPU.	Mondrian partitioning on tens of thousands of rows completes in seconds to minutes on a CPU.
Architecture and integration	ITDD-approved de-identified source copy, then prohibited-field and provenance gate, then quality profiler, then field governance classifier, then candidate release-profile generator, then equivalence-class and small-cell risk engine, then generalisation and suppression engine, then utility evaluator, then release-readiness report and dashboard	File-based import only. No connection to any city production system, which is an ethics requirement as much as an engineering choice.
Security and maintenance	Restricted external storage path outside Git and outside the thesis vault; checksum, schema version, and provenance recorded on import; prohibited-field scan before any analytical module loads data; no linkage attempt against any external source; audit logging	The no-linkage rule is enforced by design, since the system holds no external dataset to link against.
Technical expertise	Python, pandas, SQL, and Streamlit are all in hand. Mondrian and equivalence-class risk	ARX or pycanon serve as reference implementations to validate the custom Mondrian



	measurement are new to the group but are well specified in published pseudocode.	output against, so a subtle implementation error can be caught rather than defended.
Limitations and alternatives	Generalisation hierarchies cannot be finalised until the approved schema and data dictionary are seen.	Hierarchies are drafted against the public dashboard's visible fields, and the pipeline is developed and validated on a public benchmark microdata set such as the UCI Adult dataset before the real extract arrives.

Technical finding. The stack is open source and familiar, and the only genuinely new component, Mondrian, has published pseudocode and reference implementations available for cross-checking. The critical path holds no external API and no specialised hardware. The technical risk is low; the constraint is data access, treated in section 4.

4. Dataset and Data Feasibility

Requirement	Current Assessment
Dataset name and description	One ITDD-approved de-identified record-level dataset from the QC Open Data programme, most likely Helpline 122 service-request records, requested after direct-identifier removal but before dashboard aggregation, together with its data dictionary, category definitions, refresh schedule, and known quality notes.
Source, owner, and provider	Quezon City Information Technology Development Department, with the originating source department as a probable co-approver.
Public link	The QC Open Data Dashboard is publicly viewable at quezoncity.gov.ph/program/qc-open-data-dashboard/ , and the city publishes a documented Request Information service. The dashboards present aggregated views; a record-level extract is not publicly downloadable and must be requested.
Data type and file format	Expected CSV or a database export. The dashboard surface must be inspected first to record which fields, filters, periods, and aggregation levels are already public, since that inspection needs nobody's permission and can be done in Week 1.
Estimated size	The study requires at least 10,000 usable rows. Helpline 122 is a citywide complaint and service channel with dashboards covering 2019 onward, so the volume is plausible, and the exact record count will be confirmed when ITDD replies to the request already sent.
Features and attributes	Request or transaction date, service or concern category and sub-category, barangay or district, status, resolution time, and operational metadata. Narrative text, exact locations, account identifiers, and direct identifiers are excluded at source and must not be received.
Labels and target variables	Not a prediction task, so no outcome label is required. The measured quantities are equivalence-class size distribution, minimum k, proportion of records in small cells, information loss, and aggregate-query error under each candidate release profile.
Accessibility and collection	Four steps. First, verify the public surface through the browser and record what is already exposed. Second, review the exact field list, ethics route, and letter wording with the adviser. Third, submit a formal request through QC's Request Information service addressed to ITDD. Fourth, receive under agreed terms only after ethics clearance. A complete letter draft already exists in the group's working files.
Permission and licensing	The approved scope, publication conditions, and any data-sharing requirements are set by ITDD and the source department. The group does not publish the approved record-level extract under any circumstance.
Quality, completeness, and bias	Expected issues include missing or inconsistent categories, changing status codes across years, unknown refresh delays, schema changes, and reporting bias toward barangays and populations that use the channel more. That reporting bias is reported as a limitation and is not corrected for, since correcting it would require assumptions the data cannot support.
Preprocessing and storage	Immutable restricted copy with checksum and provenance; prohibited-field scan before any analytical module runs; all work on versioned derivatives; no attempt to link the research copy to any public dataset or to recover identities.
Training, validation,	Not a supervised split. Evaluation compares release profiles on the same dataset across multiple



testing	values of k , reporting risk and utility for each. The pipeline is first validated on a public benchmark microdata set with published k -anonymity results, so correctness is established before the real extract is touched.
Alternative data plan	Three fallbacks in order. First, a different QC dataset if the first choice is refused, since the method is dataset-agnostic. Second, a Philippine Statistics Authority anonymised microdata product, which is a real and obtainable microdata source. Third, a published benchmark microdata set such as UCI Adult, which preserves the full method and every measurement but loses the local government finding and the practical contribution to the city.

Dataset finding. Stronger than Plan C on this criterion, for two reasons that are both verifiable rather than hoped for: Quezon City operates a documented public request channel with a named responsible department, and a substantial part of the Week 1 work, inspecting the public dashboard surface, requires nobody's approval. The request has already been sent through an identified contact in the city, so what is outstanding is the reply and the shape of the record-level extract, and the response time is the part that stays outside the group's control.

5. Algorithm or Model Feasibility

Evaluation Area	Approach 1	Approach 2	Approach 3
Name of algorithm or approach	Fixed small-cell suppression threshold, the current common practice baseline	Mondrian multidimensional k -anonymity through greedy recursive partitioning	Differential privacy using the Laplace mechanism on aggregate queries
Purpose in the proposed project	Establish the baseline that the city's present approach approximates, so any improvement is measured against real practice rather than a straw man	Generate release profiles that reach a declared k with less information loss than the baseline	Provide a formal-guarantee comparison point for aggregate statistics
Dataset requirements	Any tabular dataset with categorical fields	Identified quasi-identifiers and a generalisation hierarchy defined per attribute	A defined query set and a privacy budget; does not release record-level data at all
Expected performance	Meets the threshold but over-suppresses, discarding detail that a smarter partition would have kept	Expected to reach the same k with measurably lower information loss, which is the study's central hypothesis	Strong formal guarantee, but not comparable on a record-level release task
Computational requirements	Trivial	Low to moderate; recursive partitioning completes in seconds to minutes at this scale	Low for the noise mechanism, but budget accounting adds design overhead
Advantages	Simple, explainable, and already understood by data owners	Multidimensional partitioning preserves more utility than fixed global rules; the mechanism is inspectable and can be shown to a panel	Provides a mathematical guarantee that k -anonymity cannot
Limitations	Blunt and lossy; cannot adapt to which combinations are actually rare	k -anonymity is vulnerable to homogeneity and background-knowledge attacks, and is not a legal guarantee of anonymity	Answers a different question, since it supports aggregate queries rather than a record-level release, so it is a reference point and not a substitute
Evaluation metrics	Minimum k achieved, suppression rate, information loss	Minimum k , equivalence-class size distribution, discernibility metric, normalised certainty penalty, and aggregate-query error against the ungeneralised copy	Epsilon spent against measured query error
Implementation	Very low	Moderate; roughly 200 lines with published pseudocode and	Moderate to high, mainly in



complexity		reference implementations available for cross-checking	budget accounting
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Recommended approach. Mondrian k-anonymity is the primary algorithm, because the research question is specifically whether a declared risk threshold can be reached with less information loss than blanket aggregation, and Mondrian is the established method that makes that comparison measurable on a record-level release. Fixed small-cell suppression is retained as the mandatory baseline, since a result is only meaningful against what data owners actually do today. Differential privacy is included in the comparison as a discussion reference and is explicitly out of scope for implementation, because it answers a different release question. k is treated as a configurable study parameter reported across multiple plausible values, never as a legal guarantee, and the final policy used in any conclusion is approved by the adviser or the data owner. The pipeline is validated against a public benchmark with published results before the city extract is used.

6. Operational Feasibility

Area	Assessment
Target users and stakeholders	ITDD staff and Database Management Division personnel as direct users; source-department data owners and the city Data Protection Officer as approving stakeholders; public open-data users as indirect beneficiaries.
Intended operating environment	An office workstation or group laptop running the application locally in a browser, used before a release decision rather than as part of any live publishing pipeline.
Availability of participants	Open and already contacted. Quezon City publishes a Request Information service and ITDD lists data extraction as a named service, so a defined channel exists, and the group has reached an identified contact through it. What is not yet known is the response time and the granularity the city will release.
User needs and benefits	A documented way to compare release options before publication, separate evidence on quality, residual risk, and utility, a repeatable record of why a particular granularity was chosen, and early identification of metadata gaps that could mislead public users.
Devices and facilities	Ordinary laptop with 8 GB RAM, a modern browser, and a restricted storage location. No procurement required.
Usability and accessibility	Risk and utility presented as a visible trade-off rather than a single pass or fail verdict, plain-language explanations of what k means, readable colour contrast, and keyboard-operable controls. Every recommendation shows the assumptions behind it.
Training and support	A short user guide, a walkthrough of the release-readiness report, and administrator documentation for adding hierarchies and datasets.
Maintenance	Generalisation hierarchies need updating as category dictionaries change, and threshold policy needs review as guidance evolves. Both are configuration rather than code.
Adoption challenges	The largest is that a system output could be mistaken for legal clearance to publish. The system therefore states in the report itself that it is decision support and does not replace the city Data Protection Officer. A second challenge is that ITDD may not have time to engage a student project; the study is designed so that the core evaluation can still be completed on a fallback dataset.

Operational finding. The operating environment is undemanding and the deliverable addresses a decision the city genuinely has to make before every release. Operational feasibility is better than Plan C because a documented request channel and a named responsible department already exist, and because no staff interviews or usability testing inside the city are required during the approved study.

7. Economic Feasibility



Expense	Estimated Cost	Purpose	Possible Alternative
Software licences	PHP 0	Python, pandas, Streamlit, SQLite, ARX or pycanon	All are open source; no paid tool is on the critical path
APIs and subscriptions	PHP 0	None required by design	Not applicable, since no external service is used
Cloud hosting and storage	PHP 0 to 600	Optional demonstration host for a public benchmark dataset only	Local demonstration; the restricted extract is never hosted
Dataset acquisition or annotation	PHP 0	Public request service carries no fee; no annotation is required	Public benchmark microdata if the request is refused
Hardware or equipment	PHP 0	Existing group laptops	School laboratory workstation if a laptop fails
Transportation and data collection	PHP 1,800	Request submission and follow-up visits to the QC Hall, estimated at three trips for two members	The online Request Information service removes most travel if it is accepted
Printing and documents	PHP 500	Formal request letter and proposal packet	Submit digitally where the city accepts it
Testing and deployment	PHP 600	Evaluation session incidentals	Remote evaluation with faculty and student testers
Maintenance	PHP 0	Handled within the study period	Not applicable during the thesis term
Contingency expenses	PHP 1,000	Follow-up visits or a change of target dataset	Reduce trips by using the online request channel
Total estimated cost	PHP 3,900 to 4,500	Working budget range	The lower bound applies if the online request channel is accepted and no hosting is used

Economic finding. The least expensive title alongside Plan C, and cheaper still if the city accepts an online request, since travel is the only material line item. Cost is not a constraint on this title.

8. Schedule Feasibility

Activity	Estimated Duration	Dependency or Control
Requirements gathering	Weeks 1 to 2	Includes public dashboard surface inspection, which needs nobody's approval and starts immediately
Literature and related-study review	Weeks 1 to 5	Independent of data access; k-anonymity literature is mature and accessible
Dataset acquisition or data collection	Weeks 1 to 5	Hard decision point at Week 5. The city request is submitted in Week 2; if no approved extract exists by Week 5, switch to PSA microdata or the public benchmark
Data cleaning, annotation, and preprocessing	Weeks 5 to 7	Depends on the received schema and dictionary
System design	Weeks 2 to 5	Generalisation hierarchies drafted against the public dashboard fields while the request is pending
Algorithm or model development	Weeks 4 to 9	Suppression baseline first, then Mondrian, validated on the public benchmark before the real extract is used
System implementation	Weeks 6 to 11	Dashboard and report generator built alongside the risk engine
Model training and validation	Weeks 9 to 12	Risk and utility measured across multiple values of k; custom Mondrian cross-checked against ARX or pycanon
Software and user testing	Weeks 11 to 14	Faculty and student testers; ITDD review only if the department is willing



Performance evaluation	Weeks 12 to 14	Mondrian against the fixed suppression baseline on identical risk thresholds
Revision and improvement	Weeks 14 to 16	Defect correction and report refinement
Documentation	Weeks 2 to 17	Maintained continuously; every assumption recorded as it is made
Deployment and final presentation	Weeks 17 to 18	Handover package only; no integration into any city system is permitted during the study

Activities that may cause delays. City response time is the largest and it is outside the group's control, and a government request may also need a second approval from the source department. Schema and dictionary quality is the second, since undefined categories block hierarchy construction. These are managed by submitting the request in Week 2 rather than later, by building and validating the entire pipeline on a public benchmark dataset so that no development work waits on the city, and by setting a hard Week 5 decision point with two named fallbacks. Because the method is dataset-agnostic, a fallback costs the group its local finding but not its system or its evaluation.

Schedule finding. The 18-week plan holds, and it is more robust than Plan C's because a public benchmark can substitute for the real extract at any point without invalidating the work already built.

9. Legal and Ethical Feasibility

Requirement	Planned Control
Data Privacy Act compliance	Republic Act No. 10173 applies. NPC Advisory Opinion No. 2017-027 states that removing direct identifiers does not automatically make data anonymous where singling out, linkability, or inference remains possible, so the research copy is treated as restricted personal data even after source-side de-identification. NPC Advisory No. 2025-02 on privacy engineering informs the system design.
Informed consent	Not applicable. No human participants are engaged and no identifiable private information is received. Consent is obtained only if usability testing records identifiable responses.
Research ethics approval	Exempt classification is targeted. Appendix I applies for formally authorised institutional data. The adviser decides whether ITDD as data custodian triggers Appendix D. Appendix H is expected to apply, because the system produces an automated release recommendation. No data is received before ethics clearance.
Confidentiality and anonymisation	Direct identifiers and source-system identity mappings are never received. Narrative text, exact locations, and account identifiers are excluded at source. High-risk first cases involving health details, PWD identities, QCitizen ID numbers, or beneficiary financial information are excluded from dataset selection entirely.
Intellectual property and copyright	All sources cited; library and dataset licences documented; the city retains ownership of its data and the group claims no redistribution right.
Licences	Every library on the stack carries a permissive open-source licence compatible with academic use. ARX and pycanon are used as reference cross-checks under their stated licences.
Data ownership and sharing restrictions	The group never publishes the approved record-level extract. Row-level files are kept outside Git and the thesis vault and are destroyed within thirty days after the final defence unless city terms require earlier.
Cybersecurity	Restricted storage path, checksum and schema-version recording on import, prohibited-field gate, credentials in environment variables, and audit logging. No external dataset is held, so no linkage is technically possible.
AI fairness, transparency, accountability, bias	No claim is made that a passing k value constitutes legal anonymity; the system is decision support and the report says so in its own text. Results are reported across multiple values of k rather than a single chosen threshold. Reporting bias toward barangays and populations that use the service channel more is disclosed as a limitation, since a release that looks safe for well-represented groups may not be for rare ones.



Possible misuse or harm	Two realistic harms. First, a system output mistaken for legal clearance to publish; controlled by explicit decision-support framing in the report itself and by the statement that it does not replace the Data Protection Officer. Second, the study documenting how to evaluate re-identification risk; controlled by never attempting re-identification, holding no external dataset to link against, and reporting only aggregate risk indicators rather than any vulnerable record.
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Legal and ethical finding. This title has the strongest legal grounding of the five, because the exact NPC guidance that governs it has been identified and quoted, and because privacy is the subject of the study rather than an obstacle to it. The genuine open items are the Appendix H determination, which is expected to apply, and the city's own publication conditions, which are unknown until ITDD responds.

10. Risk Assessment and Feasibility Conclusion

Identified Risk	Likelihood	Impact	Mitigation Strategy
Dataset is unavailable or insufficient	Low	High	Request submitted in Week 2; hard Week 5 decision point; two named fallbacks, being PSA anonymised microdata then a public benchmark microdata set. The system and the full evaluation survive both.
City response is slow or requires a second approval	High	Medium	Development is validated entirely on a public benchmark, so no build work waits on the city; the request explicitly names the source department as a possible co-approver.
Extract lacks exact dates or barangays	Medium	High	Without granular quasi-identifiers there is nothing to generalise, so this is asked directly in the request letter; if refused, the study moves to a dataset or benchmark that retains them.
Volume falls below the 10,000-record floor	Low	High	Helpline 122 is citywide with coverage from 2019, and the fallback benchmarks all exceed the floor comfortably.
Technology is difficult to implement	Low	Medium	Mondrian has published pseudocode; ARX and pycanon provide reference outputs to cross-check against, so an implementation error is caught rather than defended.
Algorithm produces poor results	Low	Medium	Comparison is against a stated baseline on identical thresholds. If Mondrian does not beat fixed suppression on this dataset, that is a genuine and reportable finding, not a failed study.
Required respondents are unavailable	Low	Low	No human participants are required for the core evaluation; faculty and student testers cover usability.
Privacy or ethical concerns arise	Low	High	No direct identifiers received, no linkage attempt, no re-identification attempt, high-risk datasets excluded from selection, Exempt-targeted design, and destruction within thirty days after the defence.
System output mistaken for legal clearance	Medium	Medium	Decision-support framing stated inside the generated report, explicit statement that it does not replace the Data Protection Officer, and results reported across multiple values of k rather than as a single verdict.
Project exceeds the budget	Low	Low	Total planned spending is under PHP 4,500, almost all of it travel, which the online request channel can remove.
Project cannot be completed on time	Low	High	The pipeline is built and validated on a public benchmark independently of the city, so the worst case costs the local finding rather than the deliverable.

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. To conclude, the technology, the cost, the schedule, and the legal grounding of this title are all sound, and it carries the most robust fallback structure among the group's titles, which follows from a method that is dataset-agnostic and can be validated on public benchmark microdata, so no development work waits on the city



answering. The group has also already sent its request through an identified contact in the Quezon City government, and the city publishes a documented request channel with a named responsible department, which together make this a real and open line rather than a hopeful one. However, the shape of the extract is still unconfirmed, which covers whether an approved record-level extract will be released, whether it keeps the exact dates and barangays the study needs, and whether a second approval from the source department will be required. The title may proceed once ITDD confirms an approved de-identified extract of at least 10,000 rows that retains the quasi-identifiers the study needs to generalise. Yet if that is not confirmed by Week 5, the group should move to PSA anonymised microdata, and only then to a public benchmark set, which preserves the method while costing the study its local finding.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	4	Familiar open-source stack; Mondrian is new to the group but has published pseudocode and reference implementations
Dataset and data feasibility	3	Documented city request channel, a named department, and a contact already engaged; extract granularity still unconfirmed
Algorithm or model feasibility	4	Mature, well-documented methods with an explicit baseline and measurable outcomes
Operational feasibility	4	Undemanding environment, a real decision need, and a contact already engaged at the city
Economic feasibility	5	Under PHP 4,500, almost entirely travel
Schedule feasibility	3	Fits 18 weeks, helped considerably by benchmark-first development
Legal and ethical feasibility	3	Strongest legal grounding of the group's titles, but Appendix H is expected to apply and city conditions are unknown
Risk manageability	4	The request is sent and the method is dataset-agnostic, so fallbacks preserve the system and the evaluation
Total score	30/40	Provisional, pending the full five-title comparison
Group's ranking	4	See the final comparison table

References

- Government Data Quality Hub. (2020). The Government Data Quality Framework. gov.uk
- LeFevre, K., DeWitt, D. J., & Ramakrishnan, R. (2006). Mondrian multidimensional k-anonymity. 22nd International Conference on Data Engineering, 25.
- National Privacy Commission. (2012). Republic Act No. 10173: Data Privacy Act of 2012. privacy.gov.ph
- National Privacy Commission. (2017). Advisory Opinion No. 2017-027: Anonymized data for marketing analytics. privacy.gov.ph
- National Privacy Commission. (2025). Advisory No. 2025-02: Guidelines on privacy engineering in systems life cycle processes. privacy.gov.ph
- Quezon City Government. (n.d.). QC Open Data Dashboard. quezoncity.gov.ph/program/qc-open-data-dashboard/
- Quezon City Government. (n.d.). Quezon City Information Technology Development Department. quezoncity.gov.ph
- Quezon City Government. (2025). How to request for public information. quezoncity.gov.ph
- Samarati, P. (2001). Protecting respondents' identities in microdata release. IEEE Transactions on Knowledge and Data Engineering, 13(6), 1010-1027.



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Sweeney, L. (2002). k-anonymity: A model for protecting privacy. International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems, 10(5), 557-570.

Reference details and the full academic literature review must be verified and completed before final submission.



TARGET TITLE 5 OF 5: PLAN E

Retail Product Demand Forecasting and Price Optimization for Small and Medium Stores Using XGBoost

1. Target Thesis Title

Retail Product Demand Forecasting and Price Optimization for Small and Medium Stores Using XGBoost

Thesis specialization: Data Science

Primary technique: Predictive Analytics, from the approved Data Science technique list

Research domain: Data Analytics, Machine Learning, Demand Forecasting, and Price Optimization

Proponents: Dawn Andrei Pamesa (Leader), Rogelio Porras, John Ashley Alba, Ralph Blanco

2. Project Overview

Background of the problem. Retail stores need to make regular decisions about product prices and inventory. Customer demand changes based on factors such as price, season, promotions, and past buying patterns. Historical sales data can reveal these patterns and help estimate future demand. Statistical and machine learning methods can study sales data and examine how different prices affect demand and revenue.

Specific problem. Small and medium retail stores often use experience and previous sales to set prices and manage inventory. These methods can overlook changes in customer demand and buying behavior. Inaccurate demand forecasts can lead to excess inventory or product shortages. Poor pricing decisions can also result in missed sales and lower revenue.

Target users and beneficiaries. Small and medium retail store owners, managers, inventory staff, and employees responsible for pricing and inventory management.

Proposed solution. The study will develop a retail analytics system using historical sales and pricing data. Machine learning models will forecast product demand using previous sales, prices, dates, seasonal patterns, and promotions when data is available. The system will study how price changes affect customer demand. It will then recommend product prices based on predicted demand and set business limits. An interactive dashboard will display sales trends, demand forecasts, and recommended prices.

Main objectives.

- Analyse historical sales and pricing patterns of retail products.
- Develop and compare machine learning models for demand forecasting.
- Measure the relationship between product prices and customer demand.
- Create a price recommendation method using predicted demand.
- Develop a dashboard for viewing sales trends, forecasts, and pricing recommendations.

Expected output. The expected output is a retail analytics and decision-support system. It will provide demand forecasts, historical sales analysis, price analysis, and recommended product prices. The dashboard will present these results through charts and reports. The study will also compare forecasting models using metrics such as MAE, RMSE, and MAPE.

Intended contribution. The study will show how data science can support pricing and inventory decisions for small and medium retail stores. Store owners can use historical sales patterns to estimate future product demand and

examine possible prices. The research will provide a practical use of machine learning, statistical analysis, and demand forecasting for retail businesses.

3. Technical and Technology Feasibility

Area	Assessment
Programming	Python will serve as the main language for data preprocessing, statistical analysis, machine learning, demand forecasting, and price recommendations. HTML, CSS, and JavaScript will support the web interface.
Libraries	Pandas and NumPy will process and prepare the data. Scikit-learn will support machine learning models. Matplotlib or Plotly will create charts and graphs. XGBoost can handle demand forecasting, and SciPy can calculate recommended prices.
Storage	MySQL will store product details, sales records, prices, inventory, promotions, forecasts, and price recommendations. The system can accept CSV files for importing historical sales data.
Services	Flask or FastAPI will connect the web interface, database, forecasting models, and pricing functions. The system will not require paid external APIs for its main features.
Platform	The project will use a web-based application that works on common desktop and mobile browsers. It can run on a local server during development and testing.
Hardware	The developers can build and test the system using standard laptops or desktop computers. The selected models can run using a modern CPU, so a dedicated GPU is not required.
Architecture	The system will follow a client-server architecture. It will contain a web interface, Python backend, MySQL database, demand forecasting module, and price recommendation module.
Security and maintenance	The system will include user authentication, role-based access, password hashing, input validation, and database backups. New sales records can be used to retrain the forecasting model when needed.
Expertise and alternatives	The project requires knowledge of Python, databases, statistics, machine learning, data preprocessing, and web development. Models such as Linear Regression, Random Forest, and XGBoost can be tested and compared using MAE, RMSE, and MAPE.

Technical finding. The stack is entirely open source and familiar, and no component requires specialised hardware, paid services, or external authorisation. This is the least technically risky of the five titles.

4. Dataset and Data Feasibility

Requirement	Current assessment
Dataset	Historical retail transaction and product pricing data containing sales, product, price, quantity sold, and date information. Inventory and promotional data will also be included when available.
Provider	Primary data may be obtained from a partner retail store. Public retail datasets from sources such as Kaggle may be used for initial model development and testing.
Links	To be finalized after selecting the primary dataset or confirming a partner retail store.
Format and volume	Data will preferably be stored in CSV or Excel format. At least one year of historical transactions is preferred to capture seasonal patterns. More observations and longer historical coverage are desirable.
Features	Date, product ID/category, selling price, quantity sold, inventory level, discount or promotion status, and product cost when available. Time-based features such as day, week, month, and holidays may also be derived.
Target	The main prediction target is product demand, represented by the quantity or units sold within a defined time period. Predicted demand will then be used as an input to the price optimization process.
Collection	Historical sales records will be requested from the participating retail store and exported from its POS, spreadsheet, or existing database. Public datasets may supplement development and



	testing when appropriate.
Rights and privacy	Permission will be obtained before using store data. Personally identifiable customer information will not be required. Store-sensitive information will be anonymized when necessary and used only for research purposes.
Quality and bias	Data will be checked for missing records, duplicates, incorrect prices, outliers, and inconsistent product information. Limited historical data or missing promotion and inventory information may affect model performance.
Preprocessing	Data cleaning, missing-value handling, duplicate removal, outlier assessment, feature engineering, categorical encoding, and time-based feature extraction will be performed before model training.
Validation split	Since the data are time-dependent, the dataset will use a chronological train-validation-test split rather than randomly mixing past and future records. Time-series cross-validation may also be used when appropriate.
Fallback	If sufficient data cannot be obtained from a partner store, a suitable public retail sales dataset will be used. The project scope may also be limited to products with enough historical sales and pricing observations.

Dataset finding. Feasibility is conditional. No partner retail store has been named and no dataset link has been finalised, so the primary data source is unconfirmed. The stated fallback is stronger than it is for the government-data titles, because public retail transaction datasets are openly downloadable and require no permission, but the group should settle the data source early rather than carry the question into development.

5. Algorithm or Model Feasibility

Evaluation	Approach 1	Approach 2	Approach 3
Name	Linear Regression	Random Forest	XGBoost
Purpose	Baseline model for predicting product demand and examining price-demand relationships	Nonlinear demand prediction using multiple retail factors	Advanced boosting model for demand forecasting
Data needs	Historical price, quantity sold, and basic product/time features	Historical sales with price, product, promotions, inventory, and time features	Larger structured dataset with sales, pricing, promotions, inventory, and engineered time features
Expected result	Simple and interpretable baseline	Strong performance with nonlinear retail patterns	Potentially higher predictive performance after tuning
Computing	Low	Low to moderate, CPU sufficient	Moderate, CPU generally sufficient
Advantages	Easy to interpret and provides a clear baseline	Handles nonlinear relationships and feature interactions	Handles complex relationships and provides feature importance
Limitations	May underfit complex demand patterns	Can become computationally heavier with many trees	Requires parameter tuning and careful leakage prevention
Metrics	MAE, RMSE, MAPE	MAE, RMSE, MAPE	MAE, RMSE, MAPE
Complexity	Low	Medium	Medium

Recommended approach. Linear Regression will serve as the baseline model and will be compared with Random Forest and XGBoost for retail demand forecasting. The models will use product prices, historical sales, promotions, inventory levels, and time-based features to predict future demand. Their performance will be measured using MAE, RMSE, and MAPE. The model that meets the study's predefined evaluation criteria will generate demand forecasts for the price optimization process. The pricing method will then test different prices based on predicted



demand, product cost, and business constraints to recommend a price that produces the highest expected profit or meets another defined goal.

6. Operational Feasibility

Area	Assessment
Users and stakeholders	Retail store owners, managers, inventory staff, and personnel responsible for pricing and sales decisions. Researchers and student evaluators may also use the system for academic evaluation.
Environment and facilities	A modern web browser, internet connection when deployed online, and a standard laptop or desktop computer. No specialized hardware is required.
Benefits	Provides product demand forecasts, sales and pricing analysis, and data-driven price recommendations. It can support better pricing and inventory decisions.
Usability	A simple dashboard will display sales trends, demand forecasts, recommended prices, and relevant product information using clear charts, labels, and filters.
Training and support	A user guide and system walkthrough will explain dashboard functions, data uploads, demand forecasts, and price recommendations. Technical documentation will support system maintenance and model retraining.
Maintenance	New sales data will be validated and added regularly. The forecasting models can be retrained as new data becomes available. Database backups, dependency updates, and model performance checks will also be performed.
Adoption challenges	Challenges may include limited historical sales data, incomplete pricing or inventory records, resistance to model-generated recommendations, and changes in customer demand. The system will present recommendations as decision support rather than automatically changing product prices.

Operational finding. The operating environment is undemanding and the dashboard is genuinely usable by store staff. Adoption depends on a participating retail store, and no store has yet agreed, so the evaluation plan should state who will test the system.

7. Economic Feasibility

Expense	Estimated cost	Purpose	Alternative
Software licenses	PHP 0	Python, MySQL, and open-source machine learning libraries	Free academic/open-source tools
APIs/subscriptions	PHP 0-500	Optional external services or deployment tools	Use free-tier services or locally stored retail data
Hosting/storage	PHP 0-1,000	Hosting the web application, database, and model	Free-tier hosting or local deployment
Dataset/annotation	PHP 0-500	Collection, preparation, and storage of historical retail sales data	Partner-store records or suitable public datasets
Hardware	PHP 0	Existing laptops or school computers for development and model training	School laboratory computers or Google Colab
Consultation/testing	PHP 0-1,000	Meetings with retail stakeholders, user testing, and system evaluation	Online meetings and local system testing
Contingency	PHP 1,000	Unexpected hosting, data collection, transportation, or	Use free services and local deployment when possible



		service expenses	
Total	PHP 1,000-4,000	Estimated project development expenses	A mostly free and open-source setup can reduce costs

Economic finding. The lowest projected cost of the five titles, at PHP 1,000 to 4,000, because the stack is open source, the hardware is already owned, and there is no fieldwork or transportation on the critical path.

8. Schedule Feasibility

Period	Activities
Weeks 1-2	Requirements gathering, literature review, dataset assessment, and research design
Weeks 2-4	Retail data acquisition, data cleaning, exploratory data analysis, and database setup
Weeks 4-6	System design, feature engineering, and Linear Regression baseline development
Weeks 6-10	Random Forest and XGBoost training, hyperparameter tuning, model evaluation, and price optimization development
Weeks 10-13	Dashboard development, system integration, software testing, usability evaluation, and model performance analysis
Weeks 14-16	System revision, documentation, deployment, final report, and presentation preparation

Activities that may cause delays. Limited or incomplete retail data, inconsistent file formats, missing product details, and lengthy model training may affect the project timeline. These risks will be addressed through early dataset evaluation, weekly progress targets, regular data backups, backup public datasets, early development of baseline models, and scheduled model training and testing.

Schedule finding. The 16-week plan is achievable and holds more slack than the titles that depend on an institutional data release, because a public dataset can be obtained on the first day if the partner store does not respond.

9. Legal and Ethical Feasibility

Topic	Assessment
Privacy and consent	The study will use historical retail sales and product data. Personal customer information will not be required. If private store data are used, permission will be obtained from the store, and sensitive business information will be anonymized when necessary.
Ethics review	The study is expected to involve minimal risk. Any surveys, interviews, or usability testing involving participants will follow the institution's research ethics requirements and informed consent procedures.
Ownership and licensing	Store-provided data will be used only with permission. Public datasets, software libraries, and other resources will follow their respective licenses and terms of use. All external sources will be properly cited.
Cybersecurity	User authentication, password hashing, access controls, input validation, and regular backups will be implemented. Sensitive store data and account credentials will not be publicly exposed.
Fairness and transparency	The study will document the features, preprocessing methods, model evaluation metrics, and limitations. Demand forecasting results will be evaluated using MAE, RMSE, and MAPE. Factors such as limited data, seasonal patterns, and unusual sales events will also be considered when interpreting results.
Misuse	Price recommendations will serve as decision-support information and will not automatically change product prices. Users will be informed that predictions and recommended prices are estimates based on historical data and may not reflect unexpected market changes.



Legal and ethical finding. Risk is low, because the study needs no personal customer information. The outstanding requirement is written permission from the partner store, plus the licence terms of any public dataset used.

10. Risk Assessment and Feasibility Conclusion

Risk	Likelihood	Impact	Mitigation
Dataset unavailable or insufficient	Medium	High	Assess data availability early. Obtain historical retail data from a partner store and maintain a suitable public retail dataset as a fallback. Limit the study to products with sufficient records if needed.
Technology difficult to implement	Low	Medium	Use familiar technologies such as Python, MySQL, Scikit-learn, XGBoost, and a simple web framework. Develop each component separately before system integration.
Algorithms produce poor results	Medium	High	Compare Linear Regression, Random Forest, and XGBoost. Perform feature engineering and hyperparameter tuning, and evaluate models using MAE, RMSE, and MAPE.
Respondents unavailable	Low	Low	Schedule usability testing early. If retail staff are unavailable, qualified evaluators may be considered according to the study's approved evaluation plan.
Privacy or ethical concern	Low	Medium	Obtain permission for private store data, exclude unnecessary personal customer information, anonymize sensitive business information, and securely store research data.
Project exceeds budget	Low	Medium	Use open-source software, free development tools, existing computers, and free-tier hosting when possible. Maintain a small contingency budget.
Project misses schedule	Medium	High	Follow weekly milestones, secure and assess the dataset early, develop the baseline model first, and prioritize the core demand forecasting and price optimization features.

Conclusion for this title: FEASIBLE WITH MODIFICATIONS.

Supporting findings. The proposed Retail Demand Forecasting and Price Optimization project is feasible in terms of technical, operational, economic, and academic requirements. It can be developed using available data science tools and standard computer hardware. The main concern is finding enough historical retail data that includes product prices, quantities sold, and transaction dates. Data on product costs, promotions, and inventory levels can



improve the study when available. The team should evaluate the dataset early and prepare a reliable public dataset as a backup. Proper data preparation, model testing, system development, and project scheduling will help the team complete the study within the planned timeframe.

Provisional Feasibility Score for This Title

Scores use the activity's 1 to 5 scale, where 5 is highly feasible. These scores were assigned by the group's assessment of the submitted write-up and should be confirmed by the author before submission. The group's ranking across all five titles is carried in the final comparison table.

Criterion	Score	Basis
Technical and technology feasibility	5	Ordinary open-source web and analytics stack, no specialised component
Dataset and data feasibility	3	No partner store or dataset link confirmed, but public retail datasets are an immediate fallback
Algorithm or model feasibility	4	Three established approaches compared on the same metrics
Operational feasibility	3	Usable deliverable, but no participating store or named evaluator yet
Economic feasibility	5	Lowest projected cost, no fieldwork or paid services
Schedule feasibility	4	16-week plan with no external authorisation on the critical path
Legal and ethical feasibility	4	No personal customer data required; store permission still needed
Risk manageability	3	Data risk is real but has a stated and immediately available fallback
Total score	31/40	See the final comparison table
Group's ranking	3	See the final comparison table

References

Ferreira, K. J., Lee, B. H. A., & Simchi-Levi, D. (2016). Analytics for an online retailer: Demand forecasting and price optimization. *Manufacturing & Service Operations Management*, 18(1), 69–88. <https://doi.org/10.1287/msom.2015.0561>

Liu, C., & Sustik, M. A. (2021). Elasticity based demand forecasting and price optimization for online retail. arXiv. <https://arxiv.org/abs/2106.08274>

Massaro, A., Panarese, A., Giannone, D., & Galiano, A. (2021). Augmented data and XGBoost improvement for sales forecasting in the large-scale retail sector. *Applied Sciences*, 11(17), 7793. <https://doi.org/10.3390/app11177793>

Inokuma, M., Ikeda, S., & Takano, Y. (2026). Estimating profitable price bounds for prescriptive price optimization. *Discover Analytics*, 4, Article 21. <https://doi.org/10.1007/s44257-026-00072-7>

XGBoost Developers. (n.d.). XGBoost parameters. XGBoost Documentation. <https://xgboost.readthedocs.io/en/latest/parameter.html>

Reference details and the full academic literature review must be verified and completed before submission.



FINAL COMPARISON OF THE FIVE TITLES

Rated 1 to 5, where 5 is highly feasible and 1 is not currently feasible

Criteria	Title 1 Plan A BAPTC	Title 2 Plan B Luzon Grid	Title 3 Plan C Tatalon	Title 4 Plan D QC ITDD	Title 5 / Plan E / Retail
Technical and technology feasibility	4	5	5	4	5
Dataset and data feasibility	4	4	3	3	3
Algorithm or model feasibility	4	5	4	4	4
Operational feasibility	4	4	3	4	3
Economic feasibility	4	5	5	5	5
Schedule feasibility	3	4	3	3	4
Legal and ethical feasibility	4	5	3	3	4
Risk manageability	4	4	3	4	3
Total score /40	31	36	29	30	31
Group's ranking	1	2	5	4	3

Reading the comparison. Plan B scores highest on every criterion that measures whether the group can start work today, because it is the only title that depends on no external organisation releasing anything. Plan A and Plan E follow at 31, then Plan D at 30, and then Plan C at 29, which is a narrow spread and shows that none of the five is unworkable. Plans A, C, and D each lose their points in the same place, which is dataset feasibility and risk manageability, and for the same underlying reason, since each of them waits on an institution to release records. In all three cases the request has already been sent through an identified contact, so what is outstanding is a reply and the shape of the records rather than a decision on whether to cooperate. Plan C sits last because it holds the least confirmed information about its records, as it is still unknown whether the barangay's files are digital and whether they reach the volume the study needs.

Why the ranking is not simply the score order. The ranking row records the group's own ordering, which is a deliberate decision rather than a copy of the total column. The scores measure how safely a title can be delivered, and they do not measure what the study contributes or whom it serves. Plan A is placed first because it is the group's committed direction following the group meeting, because it is the only title whose institutional partner has already replied, and because it addresses a Philippine market that no publicly identified study has yet modelled. Plan B holds the highest total and is placed second and kept as the backup title, precisely because that high total makes it the safest available substitute, and the switch remains genuinely available at the Week 3 gate rather than being a formality.

FINAL RECOMMENDATION

Primary title, supporting decisions, and backup title

Item	Decision
Primary recommended title	Title 1, Plan A: Highland Vegetable Price and Supply Glut Forecasting at the Benguet Agri-Pinoy Trading Center Using XGBoost
Reasons for selecting the title	First, it is the group's committed direction after the group meeting. Second, it is the only title with an identified institutional partner that has already responded: as of September 10, 2026 the group holds a verified request channel to the BAPTC General Manager. Third, it offers the strongest research contribution, since no publicly identified study has modelled price and supply-glut risk for this market. Fourth, it serves an identifiable group of beneficiaries, being Benguet farmers, rather than only demonstrating a method. The group acknowledges that



	Plan B scores higher on deliverability and retains it as the backup for that reason.
Confirmed technology and platform	Python with pandas, NumPy, statsmodels, scikit-learn, and XGBoost; SQL storage in PostgreSQL or SQLite; a browser dashboard built in Streamlit, deployed locally or on a free-tier service. No external API is on the critical path and no specialised hardware is required.
Recommended algorithm or model	XGBoost as the primary model for both price regression and glut-risk classification, with SARIMA as the interpretable seasonal baseline and Prophet as a secondary baseline. Selection is by chronological rolling-origin validation and every candidate is compared against naive and seasonal-naive forecasts.
Confirmed dataset or data source	Not yet confirmed, and this is the title's binding condition. DA Bantay Presyo bulletins are the accessible secondary source and are confirmed. BAPTC historical price and trading-volume records remain unconfirmed and are required for the full scope. A verified request channel now exists at baptc.center@bsu.edu.ph, addressed to Dr. Ruth C. Diego, General Manager.
Estimated cost and completion period	PHP 6,700 to 10,300 over an 18-week plan, subject to the course calendar and the group's actual budget. Travel to La Trinidad is the dominant cost.
Major risks and mitigation	The three material risks are unavailable primary data, weak forecast accuracy, and schedule delay. Mitigation: send the formal letter and proposal immediately; hold a Week 3 decision gate; narrow to price-only forecasting and withdraw the supply-glut claim if volume data is withheld; compare every model against naive baselines so a weak result is detected rather than presented; and prioritise a minimum viable dashboard over optional features.
Backup title	Title 2, Plan B: Short-Term Luzon Grid Power Demand and Heat-Stress Forecasting Using LightGBM and Heat Index Telemetry. It is the highest-scoring title at 36/40, requires no external permission, uses publicly downloadable data, carries no transportation cost, and handles no personal data. If the Week 3 gate on Plan A fails, the group can switch to Plan B without waiting on any institution.

Decision gate, stated plainly. Plan A proceeds only if exportable historical BAPTC price records, research-use permission, and adequate historical coverage are all confirmed by Week 3. If price records arrive without trading volume, the title narrows to price forecasting and the supply-glut claim is withdrawn. If nothing is confirmed by Week 3, the group moves to Plan B.

ATTACHMENTS AND SUPPORTING REFERENCES

Required attachments per the submission instructions

Attachment	Status
Dataset samples or links	Links provided in each title's section 4. Sample extracts cannot be attached for Plans A, C, and D until the respective data requests are approved. Plan B sources are publicly reachable and require no approval.
Technology documentation	Library and framework documentation is publicly available for every component named; all are open source.
Cost estimates	Provided as planning allowances in each title's section 7. These are not supplier quotations.
Supporting references	Listed at the end of each title's section. Reference details and the full academic literature review must be verified and completed before final submission.